

Tips from TIPS: The Informational Content of Treasury Inflation-Protected Security Prices

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Abstract

Treasury Inflation-Protected Securities (TIPS) are frequently thought of as risk-free real bonds. Using no-arbitrage term structure models, we show that TIPS yields exceeded risk-free real yields by as much as 100 basis points when TIPS were first issued and up to 300 basis points during the 2007–2008 financial crisis. This spread predominantly reflects the poorer liquidity of TIPS relative to nominal Treasury securities. Other factors, including the indexation lag and the embedded deflation protection in TIPS, play a much smaller role. Ignoring this spread also significantly distorts the informational content of TIPS break-even inflation, a widely used proxy for expected inflation.

I. Introduction

Treasury Inflation-Protected Securities (TIPS) are fixed-income securities whose coupons and principal payments are indexed to the non-seasonally-adjusted consumption price index (CPI) for all urban consumers.¹ Since its inception in 1997, the market for TIPS has grown substantially and now comprises about 8.6% of the entire Treasury debt market. The accumulated TIPS data provides a rich source of information to investors, policy makers, and researchers

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¹Sack and Elsasser (2004) provide a detailed description of the TIPS market.

alike. TIPS yields can be viewed as rough measures of risk-free real interest rates, an important determinant of the costs for financing private investment and public debt and arguably a better gauge of the stance of monetary policy than nominal interest rates. Importantly, the spread between yields on nominal Treasury securities and on TIPS of comparable maturities, the “break-even inflation rate” (BEI) or “inflation compensation,” is often used as a real-time proxy for market participants’ inflation expectations.

Despite the potential usefulness of these assets, this article presents evidence that it is essential to account for the lower liquidity of TIPS relative to their nominal counterparts when using them to measure real interest rates and inflation expectations. TIPS investors appear to demand extra compensations for holding these less liquid securities, thereby pushing up TIPS yields above the real yields that are consistent with nominal Treasury yields and pushing down TIPS BEI below its fundamental levels. Treating the TIPS BEI as a clean proxy for inflation expectation can be especially problematic because a combination of economically significant TIPS liquidity premiums and inflation risk premiums could potentially drive a notable wedge between the TIPS BEI and true inflation expectations. Indeed, we show that the early years of the TIPS market and the 2007–2008 financial crisis provide two prominent examples when poor liquidity significantly distorted the information content of TIPS prices.

Here we refer to a broad concept of illiquidity that may originate from a variety of market imperfections, such as those associated with the introduction of a new financial instrument (as in the early years of TIPS), various forms of trading costs and funding constraints (as experienced acutely during the 2007–2008 financial crisis), and the demand imbalances between nominal Treasuries and TIPS (as during flight-to-safety episodes). Regardless of the source of illiquidity, we emphasize that it is the expected future liquidity of TIPS relative to a comparable nominal Treasury security, rather than merely the current or the absolute level of liquidity, that should determine the liquidity premium embedded in the current TIPS yield. Recognizing the difficulty of capturing such a wide variety of current and expected future TIPS market conditions with a few observable measures, many of which had not been available until recently, we choose instead to model TIPS liquidity as driven by an unobserved TIPS-specific risk factor.

As a brief outline of the article, we first present model-free evidence that a significant portion of the movement in TIPS yields is not spanned by nominal Treasury yields. We then introduce a no-arbitrage asset pricing framework that jointly models nominal Treasury yields, TIPS yields, and realized inflation, allowing for a wedge between TIPS yields and risk-free real yields adjusted for the indexation lag. This wedge is modeled as a function of the TIPS-specific risk factor mentioned previously and is referred to as the “TIPS-indexed bond spread” or the “TIPS spread” for short. Having obtained estimates of the TIPS spread, we examine how they are linked to observable measures of the relative illiquidity of TIPS versus nominal Treasury securities while controlling for other technical factors, such as CPI seasonality, the embedded deflation floor in TIPS, flight-to-safety demand for nominal Treasury securities, and Federal Reserve purchases of TIPS.

We obtain three main findings: First, a standard 3-factor model that treats TIPS yields as risk-free, indexation-lag-adjusted real yields generates a poor fit of TIPS yields and BEI, as well as estimates of inflation expectations and inflation risk premiums with counterintuitive properties. In comparison, the two 4-factor models that allow for a TIPS spread generate notably smaller TIPS pricing errors, more reasonable estimates of inflation risk premiums, and estimated inflation expectations that are better aligned with survey inflation forecasts. Second, the estimated values of the TIPS spread were large ($\approx 1\%$) when TIPS were first issued, declined steadily thereafter through late 2003, and remained at relatively low levels until the 2007–2008 financial crisis, consistent with the notion that TIPS market liquidity conditions had been improving over time. Those estimates jumped to close to 3% in July 2008 during the TIPS sell-off but had largely returned to their precrisis levels by the beginning of 2010, a pattern similar to what has been documented for many other illiquid and/or risky asset prices. Finally, regression analysis shows that around 85% of the variation in our estimates of the TIPS spreads can be explained by observable measures of the liquidity conditions in the TIPS market. Other factors, including CPI seasonality and TIPS deflation floors, play a minor role.

The approach outlined previously differs from that of the other two studies of TIPS liquidities in the literature, those by Shen (2006) and Pflueger and Viceira (2016), who based their results on regression analysis of either TIPS BEI itself or the difference between TIPS BEI and survey-based measures of inflation expectations. The usage of a pricing model allows us to bring in additional information from the cross section of nominal yields and TIPS BEI and from realized inflation. In addition, unlike Chen, Liu, and Cheng (2010), who study TIPS in a multivariate Cox, Ingersoll, and Ross (CIR) (1985) framework, all models in this article are from the Gaussian essentially affine no-arbitrage term structure family that allows a flexible correlation structure between the factors and the market prices of risk. Such flexibilities are important for capturing the dynamics of bond risk premiums, as shown by Duffee (2002) and others, and for our purpose of accurately decomposing TIPS BEI into expected inflation, the inflation risk premium, and the TIPS spread.

To the best of our knowledge, this is the first article examining TIPS liquidity in a no-arbitrage framework and is therefore related to the fast-growing literature on the link between liquidity and asset returns.² Previous studies have documented that assets with similar payoffs can carry significantly different prices due to their varying degrees of liquidity.³ We add to the evidence by showing that this is also the case for TIPS and nominal Treasury securities. Our article is also related to studies of indexed bond pricing, most of which are conducted using data from

²Vayanos and Wang (2013) and Amihud, Mendelson, and Pedersen (2013) provide recent surveys of this literature.

³See, for example, Amihud and Mendelson (1986), Brennan, Huh, and Subrahmanyam (1998), and Brennan and Subrahmanyam (1996) for equities; Amihud and Mendelson (1991) and Longstaff (2004) for nominal Treasury securities; Longstaff, Mithal, and Neis (2005), Chen, Lesmond, and Wei (2007), Bao, Pan, and Wang (2011), and Dick-Nielsen, Feldhütter, and Lando (2012) for corporate bonds; Bongaerts, De Jong, and Driessen (2011) for credit default swaps; and Mancini, Ranaldo, and Wrampelmeyer (2013) for the foreign exchange market.

countries with longer histories of such bonds.⁴ Studies using TIPS and other U.S. inflation-linked assets are fairly recent and relatively few,⁵ and most of those using TIPS yields take them at their face value. In contrast, this article shows that there is a persistent liquidity premium component in TIPS yields that, when ignored, will significantly bias the results. Finally, this article is also related to the vast literature studying the behavior of real interest rates, inflation expectations, and inflation risk premiums with or without incorporating information from indexed bonds.⁶ As discussed in Section VII, our estimates of inflation expectations and inflation risk premiums are consistent with those obtained in other studies using sample periods similar to ours.

The remainder of this article is organized as follows: In Section II, we provide evidence that portions of TIPS yields and BEI are not spanned by nominal interest rates and are likely linked to the relative illiquidity of TIPS. Section III spells out the details of our no-arbitrage term structure models, including the specification of the TIPS-specific factor and the treatment of the indexation lag. Section IV describes the data and our estimation methodology, and Section V presents the main empirical results. Section VI examines the properties of the estimated TIPS spreads, showing that they account for a significant portion of the time-series variations in TIPS BEI and are mostly driven by the perspective and relative liquidity characteristics of TIPS versus nominal Treasury securities. Section VII compares our estimates of expected inflation and inflation risk premiums to those from other studies. Finally, Section VIII concludes.

II. A TIPS-Specific Factor: Simple Analysis

In this section, we present evidence that there is a component of TIPS yields that is not reflected in nominal Treasury yields and is likely related to the relative illiquidity of TIPS. This serves as the motivation for introducing a TIPS-specific factor when we model nominal and TIPS yields jointly.

A. Simple Regression Analysis

As a starting point, we regress the 10-year TIPS BEI, defined as the spread between the 10-year nominal yield and the 10-year TIPS yield, on 3-month, 2-year, and 10-year nominal yields.⁷ Standard finance theory suggests that nominal

⁴See Woodward (1990), Barr and Campbell (1997), Evans (1998), Remolona, Wickens, and Gong (1998), Risa (2001), and Joyce, Lildholdt, and Sorensen (2010) for the United Kingdom; Kandel, Ofer, and Sarig (1996) for Israel; and Hördahl and Tristani (2012) for the Euro area.

⁵This article and a contemporaneous study by Chen et al. (2010) are the first two to study TIPS in a no-arbitrage framework. More recent articles analyzing TIPS or inflation swaps include those by Chernov and Mueller (CM) (2012), Adrian and Wu (2010), Haubrich, Pennacchi, and Ritchken (HPR) (2012), Christensen, Lopez, and Rudebusch (2010), Pflueger and Viceira (2016), Fleckenstein, Longstaff, and Lustig (2014), Christensen and Gillan (2012), Fleming and Krishnan (2012), Grishchenko and Huang (GH) (2013), and Abrahams, Adrian, Crump, Moench, and Yu (AACMY) (2016).

⁶For studies not using indexed bonds, see, among others, Pennacchi (1991), Foresi, Penati, and Pennacchi (1997), Brennan, Wang, and Xia (2004), Buraschi and Jiltsov (BJ) (2005), and Ang, Bekaert, and Wei (ABW) (2008).

⁷We thank Greg Duffee for this suggestion. Results using three different nominal yields or using the first principal components of nominal yields are similar. See Section III.A for data details.

yields of any maturity, $y_{t,\tau}^N$, can be decomposed into the real yield, $y_{t,\tau}^R$; inflation expectation, $I_{t,\tau}$; and inflation risk premium, $\wp_{t,\tau}$:

$$(1) \quad y_{t,\tau}^N = y_{t,\tau}^R + I_{t,\tau} + \wp_{t,\tau}.$$

If TIPS yields accurately capture the underlying real yields, the TIPS BEI is the sum of expected inflation and the inflation risk premium, both parts of the nominal yields. A regression of TIPS BEI onto nominal yield curve factors can then be expected to generate a high R^2 . Conversely, variations in TIPS yields that are orthogonal to those in real yields could lead to a low R^2 .

We estimate the regression both in levels and in weekly differences for three sample periods: the full sample from Jan. 6, 1999, to Mar. 27, 2013; the precrisis period from Jan. 6, 1999, to July 25, 2007; and the postcrisis period from Aug. 1, 2007, to Mar. 27, 2013. As shown in Panel A of Table 1, the R^2 from the full-sample regression is merely 6% in levels and 39% in weekly differences, well below the R^2 in excess of 95% typically found when regressing one nominal yield onto other nominal yields. In the precrisis sample, the R^2 is higher for both specifications, 30% in levels and 59% in differences, indicating that the very low

TABLE 1
Factors Driving Nominal and Inflation-Indexed Bonds

Panel A of Table 1 reports adjusted R^2 s from regressions of 10-year TIPS break-even inflation (BEI) on 3-month, 2-year, and 10-year nominal Treasury yields over the full weekly sample of Jan. 6, 1999–Mar. 27, 2013; a precrisis subsample of Jan. 6, 1999–July 25, 2007; and a postcrisis subsample of Aug. 1, 2007–Mar. 27, 2013. Panel B reports cumulative percentages of variances of weekly changes in nominal yields only or nominal and indexed bond yields combined that are explained by their respective first four principal components. The in-sample pairwise correlations between the two sets of principal components are reported in Panel C. Panel D reports a quarterly regression of the difference between the 10-year inflation forecast from the Survey of Professional Forecasters (SPF) and the 10-year TIPS BEI on TIPS transaction volumes relative to those of nominal Treasuries, the difference between asset swap (ASW) spreads on on-the-run and off-the-run nominal Treasuries, and the average absolute TIPS curve-fitting errors. Ordinary least squares (OLS) standard errors are reported in parentheses.

Panel A. Regressing 10-Year BEI on Nominal Yields: Adj. R^2 (in %)

Sample	In Level	In Weekly Changes
Full sample	6.0	39.2
Precrisis	30.1	58.7
Postcrisis	5.0	26.1

Panel B. Variances Explained by Principal Components (full sample, in %)

PC	Nominal Yields Only	Nominal and TIPS Yields
1st	70.6	65.4
2nd	92.7	86.2
3rd	97.6	94.1
4th	99.2	97.6

Panel C. Correlation of Principal Components (full sample)

		Nominal and TIPS Yields			
		PC1	PC2	PC3	PC4
Nominal Yields Alone	PC1	0.95	−0.28	−0.15	0.01
	PC2	0.18	0.87	−0.47	0.02
	PC3	0.03	0.06	0.16	0.98
	PC4	0.01	0.02	0.06	−0.05

Panel D. The Role of Liquidity (full sample)

Constant	Relative TIPS Transaction Volume	Nominal On/Off ASW Spread Difference	Average Absolute TIPS Curve-Fitting Error	Adj. R^2
0.1766 (0.1531)	−0.1633 (0.0618)	0.0081 (0.0051)	0.0534 (0.0126)	60.2%

R^2 is mostly due to the postcrisis period, during which the adjusted R^2 declines to 5% in levels and 26% in differences, respectively. This evidence suggests that a large portion of variations in the 10-year BEI cannot be explained by factors driving the nominal yields, and even more so when the 2007–2008 financial crisis is included in the analysis.

B. Principal Components Analysis

We next turn to a principal component analysis (PCA) of the cross section of nominal and TIPS yields over the full sample. It is well known that three factors explain most of the movement in nominal yields. This is confirmed by Panel B of Table 1: Over 97% of the variation in the weekly changes of nominal yields can be explained by the first three principal components. However, once we add TIPS yields, at least four factors are needed to explain the same portion of the total variance. Panel C reports the correlations between the first four PCA factors extracted from nominal yields alone and those from a combination of nominal and TIPS yields. The first, second, and fourth factors constructed from all yields largely retain their interpretations as the level, slope, and curvature of the nominal yield curve, as attested by their high correlations with the first three nominal factors, respectively. However, the third PCA factor extracted from nominal and TIPS yields combined is not highly correlated with any of the nominal PCA factors.

C. One Potential Explanation: TIPS Liquidity Premium

One interpretation of the TIPS-specific factor identified previously is that it may reflect the lack of liquidity in TIPS relative to nominal Treasury securities. TIPS were introduced fairly recently and remain much less liquid than their nominal counterparts, against which the TIPS BEI is computed.⁸ Investors are therefore likely to demand extra compensation for holding TIPS, especially in their early years and during the 2007–2008 financial crisis when overall liquidity deteriorated, thereby pushing down TIPS prices and up TIPS yields. Indeed, TIPS outstanding, shown in Graph A of Figure 1, did not begin to rise substantially until 2004, around which time both TIPS transaction volumes and TIPS mutual funds, shown in Graphs B and C, also experienced significant growth.⁹ Transaction volumes declined notably at the end of 2008 and remained low through 2011 but rose again in recent years. In view of this history, it is plausible that TIPS yields contain a significant liquidity premium in their early years and again during the financial crisis.

A positive liquidity premium in TIPS yields can also help resolve an apparent inconsistency between long-term survey inflation forecasts and the 10-year TIPS BEI, plotted in Figure 2.¹⁰ The true BEI can be expected to be higher than survey inflation expectations if the inflation risk premium is on average positive, and they

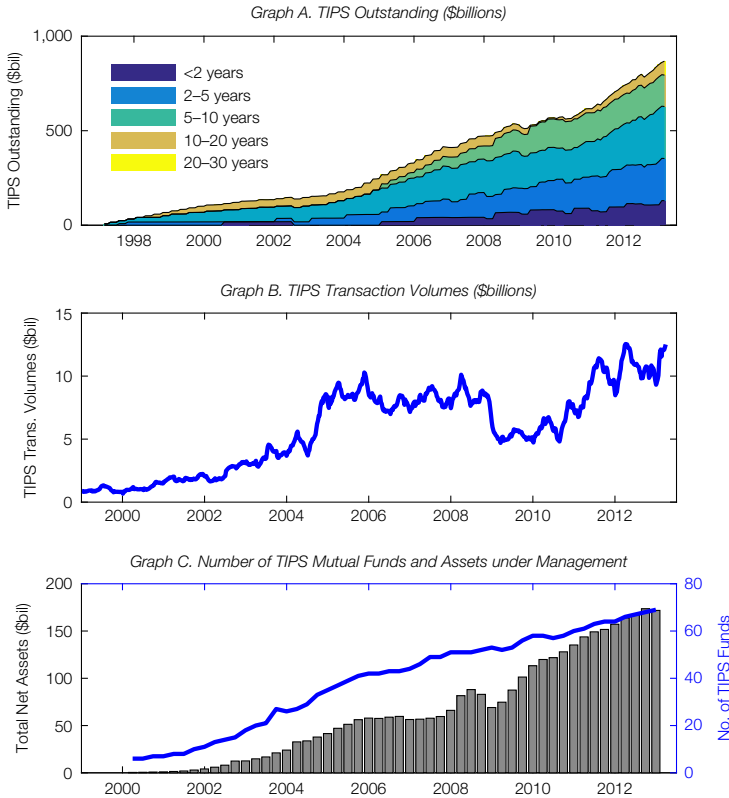
⁸Sack and Elsasser (2004) discuss liquidity conditions in the TIPS market in the early years.

⁹Data on TIPS mutual funds are from the Investment Company Institute (<http://www.ici.org>).

¹⁰We use the 10-year inflation forecasts from the Survey of Professional Forecasters (SPF) or the long-term inflation forecast from the Michigan Survey of Consumers. Mehra (2002) shows that these forecasts are unbiased, are efficient, and have predictive content for future inflation. Ang, Bekaert, and Wei (2007) and Chun (2012) show that SPF, Blue Chip, and other surveys forecast inflation better than many types of models estimated with yields only. Finally, CM (2012) show that all aforementioned survey forecasts are consistent with inflation expectations embedded in yields.

FIGURE 1
TIPS Outstanding, Transaction Volumes, and Mutual Funds

Graph A of Figure 1 plots TIPS outstanding broken down by remaining maturities, based on data reported in the Treasury's Monthly Statement of the Public Debt (MSPD). Graph B plots the weekly TIPS transaction volumes, defined as the 13-week moving average of weekly averages of daily TIPS transaction volumes reported by primary dealers in Government Securities Dealers Reports (FR-2004). Graph C plots the number of TIPS mutual funds (right axis) and the total net assets under management (left axis) from the Investment Company Institute (<http://www.ici.org>).



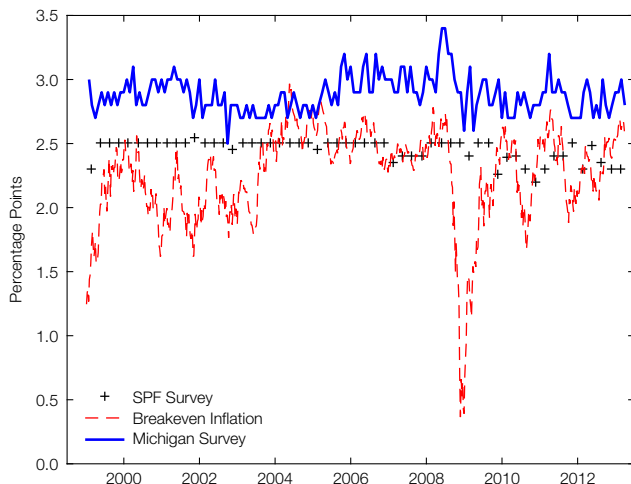
can be considered a good measure of true expected inflation if such premium is relatively small and constant over time.

However, Figure 2 shows that this is not the case: The TIPS BEI lay below survey inflation forecasts almost the entire time before 2004.¹¹ In addition, TIPS yields surged while nominal Treasury yields plummeted shortly after Lehman failed, causing the 10-year TIPS BEI to collapse by 200 basis points (bps) to below 0.5% by the end of 2008 before returning to the pre-Lehman level over the following year; throughout the same period, survey inflation expectations were practically unchanged. For inflation risk premiums to be the sole source of such

¹¹Other measures of inflation expectation based on time-series models also tend to be above the TIPS BEI in early years. Similar points are made by Shen and Corning (2001) and Shen (2006).

FIGURE 2
Survey Inflation Forecasts and TIPS BEI

Figure 2 shows the 10-year TIPS break-even inflation (BEI) (dashed line), long-horizon Michigan inflation forecast (solid line), and 10-year Survey of Professional Forecasters (SPF) inflation forecast (pluses).



notable disparities between TIPS BEI and survey forecasts, they would need to be deeply negative during both episodes and highly volatile. Economic theory does not rule out negative inflation risk premiums per se, as noted by Piazzesi and Schneider (2007) and Campbell, Sunderam, and Viceira (2017). However, as summarized by Bekaert and Wang ((2010), Table 11), most studies in the literature do find inflation risk premiums to be positive on average and relatively smooth during our sample period (see, e.g., Campbell and Shiller (1996), Foresi et al. (1997), Veronesi and Yared (1999), BJ (2005), ABW (2008), HPR (2012), CM (2012), and Hördahl and Tristani (2014)). These findings therefore cast doubt on the ability of inflation risk premiums alone to account for these significant and sometimes volatile discrepancies.

A positive TIPS liquidity premium, conversely, could push up TIPS yields and push down the TIPS BEI below the true BEI; a large enough TIPS liquidity premium could even outweigh a positive inflation risk premium and depress the TIPS BEI to levels below survey inflation forecasts. Part of the volatility of the TIPS BEI may also reflect fluctuations in liquidity premiums. Indeed, as shown in Panel D of Table 1, three proxies of the TIPS-market liquidity conditions explain about 60% of the difference between the quarterly 10-year SPF inflation forecast and the 10-year TIPS BEI over the full sample. To formally test this hypothesis, we need a framework for identifying and measuring the relevant components, including inflation expectations, inflation risk premiums, and the potential TIPS liquidity premiums. For this purpose, we now switch to the no-arbitrage term structure modeling framework and reexamine the TIPS liquidity hypothesis in Section VI.

III. A Joint Model of Nominal Yields, TIPS Yields, and Inflation

We use a no-arbitrage pricing framework that models nominal yields, TIPS yields, and inflation jointly. This approach avoids the tight assumptions that go into structural, utility-based models while still requiring the cross section of yields and inflation to be priced in an internally consistent manner that is free of arbitrage opportunities.

A. State Variable Dynamics and the Nominal Pricing Kernel

We assume that real yields, expected inflation, and nominal yields are all driven by a vector of three latent variables, $x_t = [x_{1t}, x_{2t}, x_{3t}]'$, that follows a multivariate Gaussian process,

$$(2) \quad dx_t = \mathcal{K}(\mu - x_t)dt + \Sigma dB_t,$$

where B_t is a 3×1 vector of standard Brownian motion. The nominal pricing kernel takes the form

$$(3) \quad \frac{dM_t^N}{M_t^N} = -r^N(x_t)dt - \lambda^N(x_t)' dB_t,$$

with the nominal short rate and nominal prices of risk given by

$$(4) \quad r^N(x_t) = \rho_0^N + \rho_1^{N'} x_t,$$

$$(5) \quad \lambda^N(x_t) = \lambda_0^N + \Lambda^N x_t.$$

Note that the nominal term structure in this article is fairly standard and falls into the “essentially affine” $A_0(3)$ category developed by Duffee (2002).

B. Inflation and the Real Pricing Kernel

We assume that the price-level process takes the following form:

$$(6) \quad d \ln Q_t = \pi(x_t)dt + \sigma_q' dB_t + \sigma_q^\perp dB_t^\perp,$$

where the expected natural log of the instantaneous rate of inflation, $\pi(x_t)$, is an affine function of the state variables:

$$(7) \quad \pi(x_t) = \rho_0^\pi + \rho_1^{\pi'} x_t.$$

Unexpected inflation consists of a component, $\sigma_q' dB_t$, that loads on shocks that move the nominal interest rates and expected inflation, dB_t , and a component, $\sigma_q^\perp dB_t^\perp$, that loads on an orthogonal shock $dB_t^\perp$, with $dB_t dB_t^\perp = 0_{3 \times 1}$. The orthogonal shock is included to capture short-run inflation variations that may not be spanned by yield curve movements.¹²

A real bond can be thought of as a nominal asset paying realized inflation upon maturity. Therefore, the real and the nominal pricing kernels are linked by the following no-arbitrage relation:

$$(8) \quad M_t^R = M_t^N Q_t.$$

¹²See Kim (2009) for more discussions about such variations.

As detailed in Appendix A, the real pricing kernel follows the dynamics

$$(9) \quad \frac{dM_t^R}{M_t^R} = -r^R(x_t)dt - \lambda^R(x_t)'dB_t - (\cdot)dB_t^\perp.$$

C. Nominal and Real Bond Yields

The time t prices of τ -period nominal and real bonds, $P_{t,\tau}^N$ and $P_{t,\tau}^R$, are given by

$$(10) \quad P_{t,\tau}^i = E_t \left[\frac{M_{t+\tau}^i}{M_t^i} \right], \quad i = N, R.$$

They can also be expressed in terms of expectations taken under the risk-neutral measure Q :

$$(11) \quad P_{t,\tau}^i = E_t^Q \left(\exp \left(- \int_t^{t+\tau} r_s^i ds \right) \right), \quad i = N, R.$$

Their closed-form solution can be derived following the standard literature.¹³

$$(12) \quad P_{t,\tau}^i = \exp(A_\tau^i + B_\tau^{i'}x_t), \quad i = N, R,$$

where

$$(13) \quad \frac{dA_\tau^i}{d\tau} = -\rho_0^i + B_\tau^{i'}(\mathcal{K}\mu - \Sigma\lambda_0^i) + \frac{1}{2}B_\tau^{i'}\Sigma\Sigma'B_\tau^i,$$

$$(14) \quad \frac{dB_\tau^i}{d\tau} = -\rho_1^i - (\mathcal{K} + \Sigma\Lambda^i)'B_\tau^i,$$

with initial conditions $A_0^i=0$ and $B_0^i=0_{3 \times 1}$.

Nominal and real yields therefore both take the affine form,

$$(15) \quad y_{t,\tau}^i = a_\tau^i + b_\tau^{i'}x_t, \quad i = N, R,$$

with factor loadings $a_\tau^i = -A_\tau^i/\tau$ and $b_\tau^i = -B_\tau^i/\tau$.

D. Inflation Expectations and Inflation Risk Premiums

In this model, inflation expectations also take an affine form,

$$(16) \quad I_{t,\tau} \triangleq \frac{E_t \left(\ln \left(\frac{Q_{t+\tau}}{Q_t} \right) \right)}{\tau} = a_\tau^I + b_\tau^{I'}x_t,$$

where the factor loadings a^I and b^I are given by

$$\begin{aligned} a_\tau^I &= \rho_0^\pi + (1/\tau)\rho_1^{\pi'} \int_0^\tau (I - e^{-\mathcal{K}s})\mu ds, \\ b_\tau^I &= \left(\frac{1}{\tau} \right) \int_0^\tau e^{-\mathcal{K}'s} \rho_1^{\pi'} ds. \end{aligned}$$

¹³See Duffie and Kan (1996) and Dai and Singleton (2000), among others.

From equations (15)–(16), it can be seen that the BEI, defined as before, and the inflation risk premium, defined as the difference between the BEI and the expected natural log of the inflation rate over the same horizon and denoted by $\wp_{t,\tau}$, are both affine in the state variables:

$$(17) \quad \text{BEI}_{t,\tau} \triangleq y_{t,\tau}^N - y_{t,\tau}^R = a_\tau^N - a_\tau^R + (b_\tau^N - b_\tau^R)'x_t,$$

$$(18) \quad \wp_{t,\tau} \triangleq y_{t,\tau}^N - y_{t,\tau}^R - I_{t,\tau} = a_\tau^N - a_\tau^R - a_\tau^I + (b_\tau^N - b_\tau^R - b_\tau^I)'x_t.$$

Using equation (8), we can write the price of a τ -period nominal bond as

$$(19) \quad P_{t,\tau}^N = \frac{E_t[M_{t+\tau}^R Q_{t+\tau}^{-1}]}{M_t^R Q_t^{-1}}.$$

It is then straightforward to show that the inflation risk premium $\wp_{t,\tau}$ consists of a covariance term, $c_{t,\tau}$, and a Jensen's (1906) inequality term, $J_{t,\tau}$:

$$(20) \quad \wp_{t,\tau} = c_{t,\tau} + J_{t,\tau},$$

where

$$(21) \quad c_{t,\tau} \equiv -\frac{1}{\tau} \ln \left[1 + \frac{\text{cov}_t \left(\frac{M_{t+\tau}^R}{M_t^R}, \frac{Q_t}{Q_{t+\tau}} \right)}{E_t \left(\frac{M_{t+\tau}^R}{M_t^R} \right) E_t \left(\frac{Q_t}{Q_{t+\tau}} \right)} \right].$$

In practice, the Jensen's inequality term is fairly small, and the inflation risk premium is mainly determined by the covariance between the real pricing kernel and inflation, and it can assume either a positive or a negative sign depending on how the two terms covary over time.

E. A TIPS-Specific Factor

Given the evidence presented in Section II on the existence of a TIPS-specific factor, we allow the TIPS yield to deviate from the underlying real yield. The resulting yield spread, $L_{t,\tau} = y_{t,\tau}^T - y_{t,\tau}^R$, should primarily capture the liquidity premium TIPS investors demand for holding an instrument that is less liquid than nominal Treasury securities, but it may also reflect other technical factors, such as seasonal variations in headline CPI and the embedded deflation protection in TIPS. We examine the relative importance of each of these factors in Section VI. Because the relative illiquidity of TIPS would raise TIPS yields, we would in general expect $L_{t,\tau}$ to be positive.

To model $L_{t,\tau}$, we assume that investors discount TIPS cash flows by adding a positive spread, l_s , to the instantaneous real short rate, resulting in a TIPS yield that exceeds the real yield by

$$(22) \quad L_{t,\tau} = -(1/\tau) \ln E_t^Q \left(\exp \left(- \int_t^{t+\tau} (r_s^R + l_s) ds \right) \right) - y_t^R.$$

This is analogous to the corporate bond pricing literature, where defaultable bonds are priced by discounting future cash flows using a default- and liquidity-adjusted

short rate (see Duffie and Singleton (1999), Longstaff et al. (2005), and Driessen (2005)). Without the instantaneous spread l_t in equation (22), the TIPS and the real yields y^R coincide, and $L_{t,\tau}$ becomes 0 (see equation (11)).

We assume that l_t has both a systematic component and a TIPS-specific component:

$$(23) \quad l_t = \gamma' x_t + \tilde{\gamma} \tilde{x}_t,$$

where the TIPS-specific factor, $\tilde{x}_t$, follows the Vasicek (1977) process and is independent of all other state variables contained in x_t :

$$(24) \quad d\tilde{x}_t = \tilde{\kappa}(\tilde{\mu} - \tilde{x}_t)dt + \tilde{\sigma}dW_t,$$

with $dW_t dB_t = 0_{3 \times 1}$. A nonzero γ allows the TIPS–real yield spread to be correlated with the other state variables in the economy. Finally, we assume that the idiosyncratic factor $\tilde{x}_t$ carries a market price of risk of

$$(25) \quad \tilde{\lambda}_t = \tilde{\lambda}_0 + \tilde{\lambda}_1 \tilde{x}_t.$$

Appendix B shows that the spread between TIPS and real yields takes the affine form

$$(26) \quad L_{t,\tau} = [\tilde{a}_\tau + (a_\tau^\tau - a_\tau^R)] + [(b_\tau^\tau - b_\tau^R)' \tilde{b}_\tau] \begin{bmatrix} x_t \\ \tilde{x}_t \end{bmatrix}.$$

Ignoring the indexation lag for now, the TIPS yield in this model is given by

$$(27) \quad y_{t,\tau}^\tau = y_{t,\tau}^R + L_{t,\tau}.$$

Appendix C shows that this model can be restated in a 4-factor Gaussian framework, in which the expanded state variables include x_t and the de-measured TIPS-specific factor, $\tilde{x}_t - \tilde{\mu}$.

F. Indexation Lag

An indexation lag of about 2.5 months introduces an additional complication in the pricing of TIPS. This implies that TIPS holders receive compensation for inflation over the 2.5 months prior to the purchase date but are still exposed to inflation risks during the final 2.5 months before the maturity or sale of the bond. In general, the yield on a τ -year indexed bond with an indexation lag of l years differs from the yield on a fully indexed real bond for two reasons: First, inflation between the two l -year periods could diverge, and second, investors would demand a risk premium for bearing inflation risks over the final l -year period:

$$(28) \quad y_{t,\tau,l}^l = y_{t,\tau}^R + \frac{1}{\tau} \left[E_t \ln \frac{Q_{t+\tau}}{Q_{t+\tau-l}} - \ln \frac{Q_t}{Q_{t-l}} \right] + \wp_{t,\tau,l}^{\text{IL}},$$

where $\wp_{t,\tau,l}^{\text{IL}}$ denotes the indexation-lag premium.

Evans (1998) and Risa (2001) have found the indexation-lag premium to be small, between 1.5 and 6 bps, for U.K. inflation-linked gilts with an even longer

indexation lag of 8 months. Nonetheless, the indexation-lag effect on yields could be large during periods with significantly above- or below-trend inflation. For example, annualized CPI inflation was running below -9% in each of the last 3 months of 2008; over the same period, TIPS liquidity also reportedly deteriorated rapidly. It is therefore important to explicitly account for the indexation lag.

To do so, we follow Risa (2001) and note that at time t , a τ -year indexed bond with a l -year indexation lag is a claim to a nominal payoff of Q_{T-l}/Q_{t-l} , to be received at time $T = t + \tau$. Assuming that the current price level Q_t is observed without error at time t , we first consider an artificial indexed bond that pays Q_{T-l}/Q_t at time T . Let $y'_{t,\tau,l}$ and $\tilde{y}'_{t,\tau,l}$ represent the yields on the actual and the artificial indexed bond, respectively. The relationship between the two yields is

$$(29) \quad y'_{t,\tau,l} = -\frac{1}{\tau} \ln E_t \left[\frac{M_T^N}{M_t^N} \frac{Q_{T-l}}{Q_{t-l}} \right] = -\frac{1}{\tau} \ln E_t \left[\frac{M_T^N}{M_t^N} \frac{Q_{T-l}}{Q_t} \right] - \frac{1}{\tau} \ln \frac{Q_t}{Q_{t-l}} \\ = \tilde{y}'_{t,\tau,l} - \frac{1}{\tau} \ln \frac{Q_t}{Q_{t-l}}.$$

This implies that to derive the yield on the *actual* indexed bond, we can first calculate the yield on the *artificial* indexed bond, $\tilde{y}'_{t,\tau,l}$, similarly to how we price the real bond, and then subtract the realized inflation over the previous 2.5 months. Finally, to incorporate the TIPS-specific factor, the cash flows of both indexed bonds are discounted, taking into account the instantaneous TIPS spread, l_t , specified in equation (23). Appendix D describes these steps in more details.

G. A Comparison with Previous Studies

Some of the models studied in the earlier literature (e.g., Pennacchi (1991), Campbell and Viceira (2001)) can be viewed as special cases of this model. For example, Pennacchi's (1991) model corresponds to a 2-factor version of our model with constant market price of risk. Campbell and Viceira (2001) is also a special case of this model because their real term structure has a lower dimension than the nominal term structure. In this article, we let the data decide the dimensionality of the real term structure.

Overall, compared with previous studies, two main features of this model help us better distinguish the inflation risk premium and the liquidity premium components of the TIPS BEI. On the one hand, the use of price-level data Q_t in the estimation and the unrestricted correlation structure between factor innovations help us better pin down expected inflation and the inflation risk premium. On the other hand, the higher dimensionality of the real term structure, the estimation of which is assisted by the additional information from TIPS yields, allows us to better identify the parameters governing the real yield dynamics. As a result, the spread between TIPS and indexation-lag-adjusted real yields is pinned down and can be estimated separately from the inflation risk premium. These features can only be fully appreciated when considered within the context of the empirical methodology used to estimate the model, to which we now turn.

IV. Data and Empirical Methodology

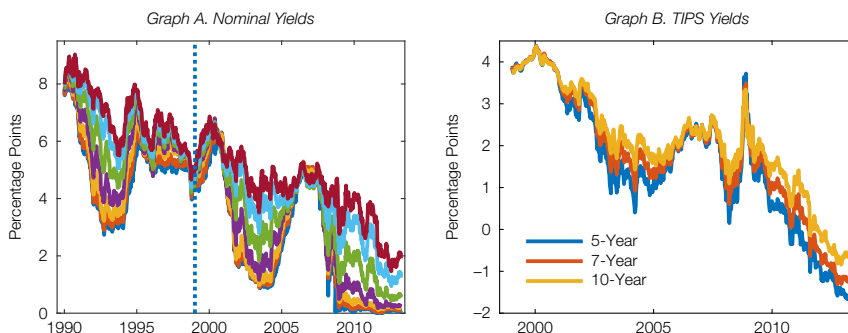
A. Data

We use 3- and 6-month, 1-, 2-, 4-, 7-, and 10-year nominal yields and CPI-U data from Jan. 1990 to Mar. 2013. In contrast, our TIPS yields are restricted by data availability and cover a shorter period from Jan. 1999 to Mar. 2013, with the earlier period without TIPS data (1990–1998) treated as missing observations.¹⁴ We sample yields at the weekly frequency and assume that the monthly CPI-U data are observed in the last week of the current month.¹⁵ Shorter-maturity TIPS yields cannot be estimated reliably before 2002 because there was only one TIPS with maturity below 5 years. We therefore only use 5-, 7-, and 10-year TIPS yields in our estimation. All nominal and TIPS yields used in the estimation are plotted in Figure 3. Although TIPS are indexed to non-seasonally-adjusted CPI, we use seasonally adjusted CPI inflation because the models we estimate do not accommodate seasonality; this is not expected to have a big effect due to the relatively long maturities of our TIPS yields.

The sample period 1990–2013 was chosen as a compromise between utilizing more data to improve the efficiency of estimation and having a more homogeneous sample with no structural breaks in the relation between term-structure variables and inflation.¹⁶ Further, to avoid running into small-sample problems, we follow Kim and Orphanides (2012) and supplement the data with survey

FIGURE 3
Nominal and TIPS Yields

Graph A of Figure 3 plots the 3- and 6-month and 1-, 2-, 4-, 7-, and 10-year nominal yields, with the vertical dashed line denoting Jan. 6, 1999, the date when TIPS data became available. Graph B plots the 5-, 7-, and 10-year TIPS yields.



¹⁴The 3- and 6-month T-bill yields are from the Federal Reserve Board's H.15 release and are converted to a continuously compounded basis. Longer-term nominal yields and TIPS yields are based on zero-coupon yield curves fitted at the Federal Reserve Board. In particular, nominal yields are based on the Svensson (1995) curve specification for the entire sample; TIPS yields are based on the Nelson and Siegel (1987) curve specification prior to Jan. 2004 and the Svensson (1995) curve specification thereafter. See Gürkaynak, Sack, and Wright (2007), (2010) for details.

¹⁵Here we abstract from the issue of real-time data by assuming that investors correctly infer the current inflation rate in a timely fashion.

¹⁶The 1979–1983 episode of the Federal Reserve's experiment with reserve targeting is a well-known example.

forecasts of short-term rates to help stabilize the estimation and better pin down some of the model parameters. Specifically, we use the 6- and 12-month-ahead forecasts of the 3-month T-bill rate from Blue Chip Financial Forecasts, which are available monthly, and allow the size of the measurement errors to be determined within the estimation. We also use the semiannual long-range (5–10 years ahead) forecast of the same rate, with the standard deviation of its measurement error fixed at a fairly large value of 75 bps at an annual rate. This is done to prevent the long-horizon survey forecasts from having an unduly strong influence on the estimation, similar to a quasi-informative prior in a Bayesian estimation.

Finally, in most cases, we include median SPF forecasts of average inflation over the following year and over the next 10 years as additional data inputs to model estimation. When included, those survey forecasts are also treated as noisy measures of their model counterparts.

B. Identification and Summary of Models

We only impose restrictions that are necessary for achieving identification to allow a maximally flexible correlation structure between the factors, which has shown to be critical in fitting the rich behavior of risk premiums observed in the data (see Duffie and Kan (1996), Dai and Singleton (2000)). In particular, we employ the following normalization:

$$(30) \quad \begin{aligned} \mu &= 0_{3 \times 1}, \quad \Sigma = \begin{bmatrix} 0.01 & 0 & 0 \\ \Sigma_{21} & 0.01 & 0 \\ \Sigma_{31} & \Sigma_{32} & 0.01 \end{bmatrix}, \\ \mathcal{K} &= \begin{bmatrix} \mathcal{K}_{11} & 0 & 0 \\ 0 & \mathcal{K}_{22} & 0 \\ 0 & 0 & \mathcal{K}_{33} \end{bmatrix}, \quad \tilde{\sigma} = 0.01, \end{aligned}$$

and we leave all other parameters unrestricted. It can be shown that any specification of the affine Gaussian model that has a $\mathcal{K}$ matrix with all-real eigenvalues can be transformed to this form.¹⁷

To summarize, we consider three model specifications that differ in how TIPS yields are modeled, including one model that equates TIPS yields with indexation-lag-adjusted real yields (Model NL),¹⁸ one model that allows TIPS yields to differ from indexation-lag-adjusted real yields by a spread that is driven only by the TIPS-specific factor (Model LI), and the full model allowing the TIPS spread to also be correlated with other state variables in the economy (Model LII). Table 2 summarizes these model specifications and the associated parameter restrictions. Among these models, Models NL and LI can both be considered special cases of Model LII. In addition, Model NL has a 3-factor representation of TIPS

¹⁷With the normalization in equation (30), the specification we estimate in this article can be shown to be equivalent to that of Sangvinatsos and Wachter (2005). The main difference between their article and ours is empirical: They use a much longer sample, which would be desirable if the relationship between inflation and interest rates can be assumed to be stable.

¹⁸This is similar to the model studied by Chen et al. (2010), who also use a CIR-type model, which is known to have problems in fitting risk premiums. Their model also implies nonnegative instantaneous inflation expectation and precludes the possibility of deflation.

TABLE 2
Summary of Models

Table 2 lists the parameter restrictions placed on the three models we estimate, including a model assuming zero TIPS liquidity premiums (Model NL), a model assuming TIPS liquidity premiums that are orthogonal to the other state variables in the economy (Model LI), and a model that allows correlation between TIPS liquidity premiums and the other state variables. Models estimated without using survey forecasts of inflation are denoted by an “-NoIE” suffix.

Models	Restrictions and Identifications
NL/NL-NoIE	$\gamma = 0_{3 \times 1}$, $\tilde{\gamma} = 0$, $\tilde{\kappa}$, $\tilde{\mu}$, $\tilde{\lambda}_0$, $\tilde{\lambda}_1$ unidentified
LI	$\gamma = 0_{3 \times 1}$, $\tilde{\gamma}$, $\tilde{\kappa}$, $\tilde{\mu}$, $\tilde{\lambda}_0$, $\tilde{\lambda}_1$ unrestricted
LII	γ , $\tilde{\gamma}$, $\tilde{\kappa}$, $\tilde{\mu}$, $\tilde{\lambda}_0$, $\tilde{\lambda}_1$ unrestricted

yields, whereas in the other two models, TIPS yields have a 4-factor specification. We estimate all three models using survey forecasts of inflation as additional data inputs. For comparison purposes, we also estimate Model NL without survey inflation forecasts and denote it with a “-NoIE” suffix.

C. Estimation Methodology

We rewrite the model in a state-space form and estimate it by maximum likelihood using the Kalman filter. More details are provided in Appendix E. Two aspects are worth noting here: First, the natural log of the price level $q_t \equiv \ln Q_t$ is nonstationary, so we use a diffuse prior for q_t when initializing the Kalman filter. Second, inflation, survey forecasts, and TIPS yields are not available for all dates, which introduces missing data in the observation equation, and are handled in the standard way by allowing the dimensions of the matrices A and B in equation (E-4) to be time-dependent (see, e.g., Harvey (1989), Section 3.4.7). To facilitate the estimation and also to make the results easily replicable, we follow a few easy steps in estimating all models:

- 1) We first perform a “pre-estimation,” where a set of preliminary estimates of the parameters governing the nominal term structure is obtained using nominal yields and survey forecasts of the 3-month T-bill rate alone.
- 2) Based on these estimates and data on nominal yields, we can obtain a preliminary estimate of the state variables, x_t .
- 3) A regression of the monthly inflation onto estimates of x_t obtained in the second step gives a preliminary set of estimates of the parameters governing the inflation dynamics.
- 4) Finally, these preliminary estimates are used as starting values in the full, 1-step estimation of all model parameters.

V. Empirical Results

A. Parameter Estimates and Overall Fit

1. Parameter Estimates

Table 3 reports selected parameter estimates for all four models.¹⁹ We draw four main conclusions: First, parameters governing the nominal term structure are

¹⁹Complete parameter estimates can be found in the Internet Appendix (available at www.jfqa.org).

TABLE 3
Selected Parameter Estimates

Table 3 reports selected parameter estimates and standard errors for all three models we estimate. Standard errors are calculated using the Berndt, Hall, Hall, and Hausman (BH-H) (1974) formula and are reported in parentheses. Complete parameter estimates can be found in the Internet Appendix.

	Model NL-NoIE		Model NL		Model LI		Model LII	
<i>Panel A. State Variable Dynamics: $dx_t = \mathcal{K}(\mu - x_t)dt + \Sigma dB_t$</i>								
$\mathcal{K}_{11}$	0.8550	(0.3533)	0.6849	(0.4589)	0.8302	(0.6993)	0.4317	(0.1622)
$\mathcal{K}_{22}$	0.1343	(0.0562)	0.1309	(0.0471)	0.1004	(0.0425)	0.0961	(0.0499)
$\mathcal{K}_{33}$	1.4504	(0.3633)	1.4259	(0.7216)	1.2353	(0.9516)	1.8425	(0.4757)
$100 \times \Sigma_{21}$	-0.7526	(0.5524)	-1.8236	(1.1939)	-1.1547	(0.8448)	-1.6133	(0.9020)
$100 \times \Sigma_{31}$	-4.4450	(4.8007)	-4.8415	(8.4964)	-7.1258	(30.8741)	-1.7824	(0.9339)
$100 \times \Sigma_{32}$	-0.9597	(0.2356)	-1.0313	(0.2948)	-1.0456	(0.4755)	-0.7864	(0.1713)
<i>Panel B. Natural Log of the Price Level: $d \ln Q_t = \pi(x_t)dt + \sigma'_q dB_t + \sigma_q^{\perp} dB_t^{\perp}$, $\pi(x_t) = \rho_0^{\pi} + \rho_1^{\pi'} x_t$</i>								
ρ_0^{π}	0.0262	(0.0016)	0.0285	(0.0015)	0.0294	(0.0021)	0.0288	(0.0026)
$\rho_1^{\pi'}$	-0.0326	(0.5805)	-0.4711	(1.7446)	-0.5261	(4.3530)	0.1582	(0.3076)
$\rho_{1,2}^{\pi'}$	0.0867	(0.0578)	0.2378	(0.0400)	0.3515	(0.0849)	0.2684	(0.0300)
$\rho_{1,3}^{\pi'}$	-0.2213	(0.1859)	-0.2804	(0.1584)	-0.1999	(0.2596)	-0.1356	(0.1442)
$100 \times \sigma_{q,1}$	-0.0796	(0.0445)	0.0038	(0.0734)	0.0000	(0.1009)	-0.1495	(0.0409)
$100 \times \sigma_{q,2}$	0.0066	(0.0673)	0.0869	(0.0739)	0.1625	(0.0620)	0.0763	(0.0581)
$100 \times \sigma_{q,3}$	-0.0278	(0.0589)	-0.2586	(0.0459)	-0.1526	(0.0674)	0.0224	(0.0619)
$100 \times \sigma_q^{\perp}$	0.9229	(0.0268)	0.9461	(0.0300)	0.9508	(0.0346)	0.8975	(0.0264)
<i>Panel C. TIPS Liquidity Premium: $l_t = \tilde{\gamma}\tilde{x}_t + \gamma'x_t$, $d\tilde{x}_t = \tilde{\kappa}(\tilde{\mu} - \tilde{x}_t)dt + \tilde{\sigma}dW_t$, $\tilde{\lambda}_t = \tilde{\lambda}_0 + \tilde{\lambda}_1\tilde{x}_t$</i>								
$\tilde{\gamma}$					0.8376	(0.0224)	0.8393	(0.0225)
$\tilde{\kappa}$					0.5097	(0.2113)	0.4900	(0.2051)
$\tilde{\mu}$					0.0067	(0.0049)	0.0077	(0.0050)
$\tilde{\lambda}_0$					0.3754	(0.3571)	0.4136	(0.3413)
$\tilde{\sigma}\tilde{\lambda}_1$					-0.3981	(0.2114)	-0.3770	(0.2052)
γ_1							-0.8403	(0.2826)
γ_2							-0.0527	(0.1024)
γ_3							0.0121	(0.2293)
<i>Panel D. Measurement Errors: TIPS Yields</i>								
$100 \times \delta \mathcal{T}_{.5y}$	0.5374	(0.0801)	0.5400	(0.0785)	0.0806	(0.0033)	0.0812	(0.0033)
$100 \times \delta \mathcal{T}_{.7y}$	0.4217	(0.0849)	0.4231	(0.0843)	-0.0000	(6,302.1210)	-0.0000	(6,307.8897)
$100 \times \delta \mathcal{T}_{.10y}$	0.3879	(0.0632)	0.3874	(0.0605)	0.0653	(0.0033)	-0.0644	(0.0033)
<i>Panel E. Measurement Errors: Survey Forecasts of Nominal Short Rate</i>								
$100 \times \delta f_{.6m}$	0.1890	(0.0146)	0.1893	(0.0146)	0.1872	(0.0141)	0.1891	(0.0146)
$100 \times \delta f_{.12m}$	0.2965	(0.0222)	0.2945	(0.0218)	0.2944	(0.0219)	0.2968	(0.0224)

fairly robustly estimated and are almost identical across all models. All estimations uncover a factor that is fairly persistent with a half-life of about 5 years. All four models also exhibit a similar fit to nominal yields and survey forecasts of nominal short-term interest rates, generating fitting errors at or below 6 bps for most maturities and slightly larger at around 13 bps for the 3-month yield.²⁰

Second, there are notable differences in the estimates of parameters governing the expected inflation process. In particular, the loading of the instantaneous inflation on the second and the most persistent factor, $\rho_{1,2}^\pi$, is negligible in Model NL-NoIE but becomes larger and statistically significant when survey inflation forecasts are used in the estimation. As a result, the monthly autocorrelation of 1-year-ahead inflation expectation is about 0.9 in Model NL-NoIE but above 0.99 in all other models. As we will see later, the lack of persistence in the inflation expectation process prevents Model NL-NoIE from generating meaningful variations in longer-term inflation expectations as we observe in the data.

²⁰Selected fitting errors for nominal and TIPS yields are shown in the Internet Appendix.

Third, the fit to TIPS yields is significantly better in models with a TIPS-specific factor. For example, the fitting errors on the 10-year TIPS yield are around 40 bps in both Model NL-NoIE and Model NL, but they are much smaller at around 7 bps in Models LI and LII. Some of the fitting errors are found to have substantial serial correlations. For example, in the case of the 7-year TIPS, we obtain a weekly AR(1) coefficient of 0.94 in all models. The finding of serial correlation in term structure fitting errors is, however, a fairly common phenomenon and has been noted by Chen and Scott (1993) and others.

Finally, parameter estimates for the TIPS-specific factor process are statistically significant in both Models LI and LII and assume similar values. The price of risk associated with this factor depends negatively on the factor itself, as can be seen from the negative $\tilde{\lambda}_1$. One possible interpretation is that the same amount of liquidity risk carries higher risk premiums when liquidity is poor, which is intuitive because one would generally expect any deterioration of liquidity conditions to occur during bad economic times. In Model LII, the loadings of the instantaneous TIPS spread on the other state variables, γ , are only significant for the first factor; however, a likelihood-ratio test shows that they are jointly significant.

2. Overall Fit

Panel A of Table 4 reports some test statistics that compare the overall fit of the four models. We first report two information criteria commonly used for model selection, the Akaike information criterion (AIC) and the Bayesian information criterion (BIC). Both information criteria are minimized by the most general model, Model LII.

We also report results from likelihood-ratio (LR) tests of the two restricted models, Models NL and LI, against their more general counterparts, Model LI and LII, respectively. When testing Model NL against Model LI, we follow Garcia and Perron (1996) and calculate a conservative upper bound for the significance of the LR test statistic as suggested by Davies (1987), outlined in more detail in the Internet Appendix.²¹ This procedure overwhelmingly rejects Model NL in favor of Model LI (with an LR p -value of essentially 0). The LR test of Model LI against Model LII, however, is fairly standard. We obtain an LR statistic of 143.13 and are able to reject Model LI in favor of Model LII at the 1% level based on a χ^2_3 distribution.

B. Fitting TIPS Yields and TIPS BEI

The estimated standard deviations of TIPS measurement errors reported in Table 3 suggest that Model NL, estimated either with or without survey inflation forecasts, has trouble fitting TIPS yields. A better understanding of the problem can be gained by comparing the three rows of Figure 4, which plot the actual and model-implied TIPS yields and TIPS BEI, as well as model-implied

²¹The standard LR test does not apply here because the nuisance parameters, $\tilde{\kappa}$, $\tilde{\mu}$, $\tilde{\lambda}_0$, and $\tilde{\lambda}_1$, are not identified under the null (Model NL) and appear only under the alternative (Model LI). For discussions on testing with nuisance parameters, see, for example, Davies (1977), (1987), (2002) and Andrews and Ploberger (1994), (1995).

TABLE 4
Specification Tests

Table 4 reports various diagnostic statistics for the three models estimated. Panel A reports the number of parameters, the log likelihood, the Akaike information criterion (AIC), the Bayesian information criterion (BIC) values, and the p -value from a likelihood-ratio (LR) test of the current model against the more general model to its right, where the p -value reported for Model NL is the Davies (1987) upper bound. Panels B–D report three goodness-of-fit statistics for the 5-, 7-, and 10-year TIPS yields; 7- and 10-year TIPS break-even inflation (BEI); and 1- and 10-year survey inflation forecasts, respectively, including the correlation between the fitted series and the data counterpart (CORR), the root mean squared errors (RMSEs), and the coefficient of determination (R^2).

	Model			
	NL-NoIE	NL	LI	LII
<i>Panel A. Overall Model Fit</i>				
No. of parameters	42	44	49	52
Log likelihood	73,092.74	73,890.53	77,808.18	77,879.74
AIC	-146,101.48	-147,693.06	-155,518.35	-155,655.48
BIC	-145,887.25	-147,468.62	-155,268.41	-155,390.23
LR p -value		0.00	0.00	
<i>Panel B. Fitting TIPS Yields</i>				
<i>5-Year</i>				
CORR (in %)	94.86	94.81	99.89	99.89
RMSE	0.54	0.55	0.16	0.16
R^2 (in %)	87.51	87.42	98.90	98.91
<i>7-Year</i>				
CORR (in %)	95.71	95.70	100.00	100.00
RMSE	0.44	0.44	0.13	0.13
R^2 (in %)	90.53	90.48	99.16	99.19
<i>10-Year</i>				
CORR (in %)	96.11	96.13	99.87	99.88
RMSE	0.41	0.41	0.14	0.14
R^2 (in %)	89.20	89.18	98.80	98.83
<i>Panel C. Fitting TIPS BEI</i>				
<i>7-Year</i>				
CORR (in %)	57.13	56.93	100.00	100.00
RMSE	0.43	0.43	0.11	0.11
R^2 (in %)	26.21	25.62	95.24	95.42
<i>10-Year</i>				
CORR (in %)	39.99	40.05	98.68	98.71
RMSE	0.37	0.37	0.12	0.12
R^2 (in %)	-1.85	-1.84	89.94	90.25
<i>Panel D. Matching Survey Inflation Forecasts</i>				
<i>1-Year</i>				
CORR (in %)	38.19	82.47	80.71	85.35
RMSE	0.69	0.38	0.44	0.35
R^2 (in %)	-8.30	67.96	56.20	72.29
<i>10-Year</i>				
CORR (in %)	59.96	82.72	83.11	83.01
RMSE	0.53	0.31	0.30	0.30
R^2 (in %)	2.26	66.30	68.80	67.73

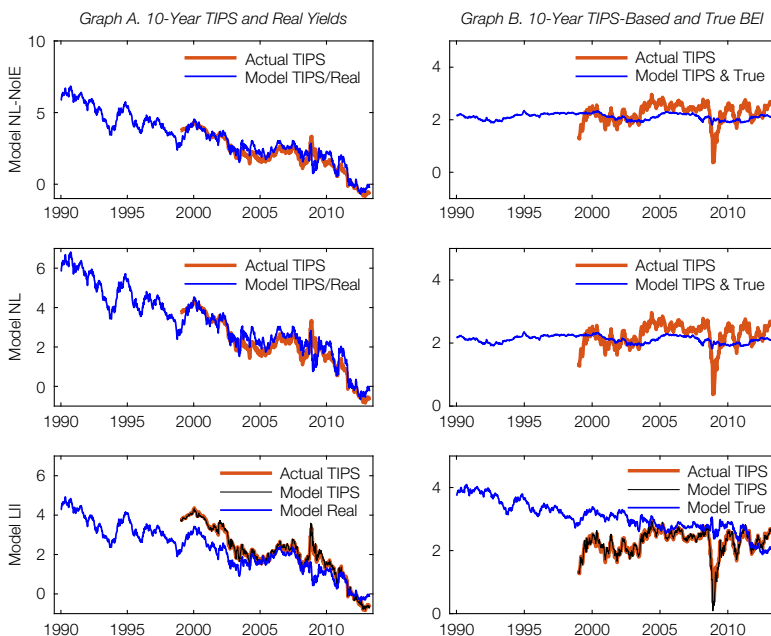
risk-free real yields and BEI, for Models NL-NoIE, NL, and LII, respectively.²² By construction, the model-implied TIPS yields (TIPS BEI) and the model-implied risk-free real yields (BEI) coincide under Model NL(-NoIE), after adjusting for the indexation lag.

The top and middle panels of Graph A of Figure 4 show that both versions of Model NL interpret the decline in 10-year TIPS yields from 1999 to 2004 as part

²² Model-implied true break-evens are calculated as the difference between model-implied nominal yields and model-implied real yields of comparable maturities. Model-implied values are calculated using smoothed estimates of the state variables. Results for Model LI are broadly similar to those for Model LII and are reported in the Internet Appendix.

FIGURE 4
Model Fit of TIPS Yields and BEI

The three rows of Figure 4 plot results from Model NL-NoIE, Model NL, and Model LII, respectively. Graph A plots the 10-year actual TIPS yields (red), the 10-year model-implied TIPS yields (black), and the 10-year model-implied real yields (blue). Graph B plots the 10-year actual TIPS break-even inflation (BEI) (red), the 10-year model-implied TIPS BEI (black), and the 10-year model-implied true BEI (blue). All numbers are in percentage points.



of a broad downward shift in real yields from 7% in the early 1990s to about 2% around 2003. The decline of less than 5% in the 10-year nominal yield over the same period is therefore attributed almost entirely to a lower real yield, leaving little room for lower inflation expectations or risk premiums. However, the literature generally finds that long-term inflation expectations likely have edged down over this period (see, e.g., Kozicki and Tinsley (2012)), and it is hard to imagine economic mechanisms that would generate such a large decline in long-term real yields. Furthermore, although the two NL models match the general trend of TIPS yields, they both have problems fitting the time variations, frequently resulting in large fitting errors, especially in the early part of the sample and again during the 2007–2008 financial crisis. In contrast, the bottom panel in Graph A of Figure 4 shows a less pronounced and more gradual decline in real yields based on Model LII, which is able to fit TIPS yields almost perfectly, as shown by the juxtaposition of the red and black lines.

In addition, the top and middle panels of Graph B in Figure 4 show that the 10-year BEI implied by the two NL models, which by construction should equal the 10-year TIPS BEI after adjusting for the indexation lag, appears too smooth compared with the actual data and misses most of its short-run variations. The poor fitting of the TIPS BEI highlights the difficulty that the 3-factor model has

in fitting nominal and TIPS yields simultaneously. In contrast, the 10-year BEI implied by Model LII, shown in the bottom panel of Graph B, exhibits substantial variations that closely match those of the actual TIPS BEI. In particular, the model-implied and the TIPS-based BEI plunge toward the end of 2008 following the Lehman collapse, consistent with reports of substantial liquidation of TIPS holdings over this period.²³

To quantify the improvement in terms of the model fit, Panels B and C of Table 4 provide three goodness-of-fit statistics for TIPS yields at 5-, 7-, and 10-year maturities and TIPS BEI at 7- and 10-year maturities, respectively. The first statistic, CORR, is the simple sample correlation between the fitted series and its data counterpart. The next two statistics, the root mean squared error (RMSE) and the coefficient of determination (R^2), are based on 1-step-ahead model prediction errors from the Kalman filter and are designed to capture how well each model can explain the data without resorting to large exogenous shocks or measurement errors.²⁴ All three statistics suggest that including a TIPS-specific factor improves the model fit for raw TIPS yields and even more so for TIPS BEI.

C. Matching Survey Inflation Forecasts

Next, we briefly examine how closely the model-implied inflation expectations mimic survey-based counterparts. Ang et al. (2007) provide evidence that survey inflation forecasts outperform various other measures of inflation expectations in predicting future inflation. In addition, survey inflation forecasts have the benefit of being a real-time, model-free measure, and hence they are not subject to model estimation errors or look-ahead biases.

Panel D of Table 4 reports the three goodness-of-fit statistics, CORR, RMSE, and R^2 , for 1- and 10-year ahead SPF inflation forecasts. When survey inflation forecasts are not used in the estimation, Model NL-NoIE generates inflation expectations that depart significantly from survey inflation forecasts, as can be seen from the large RMSEs and small and even negative R^2 statistics.

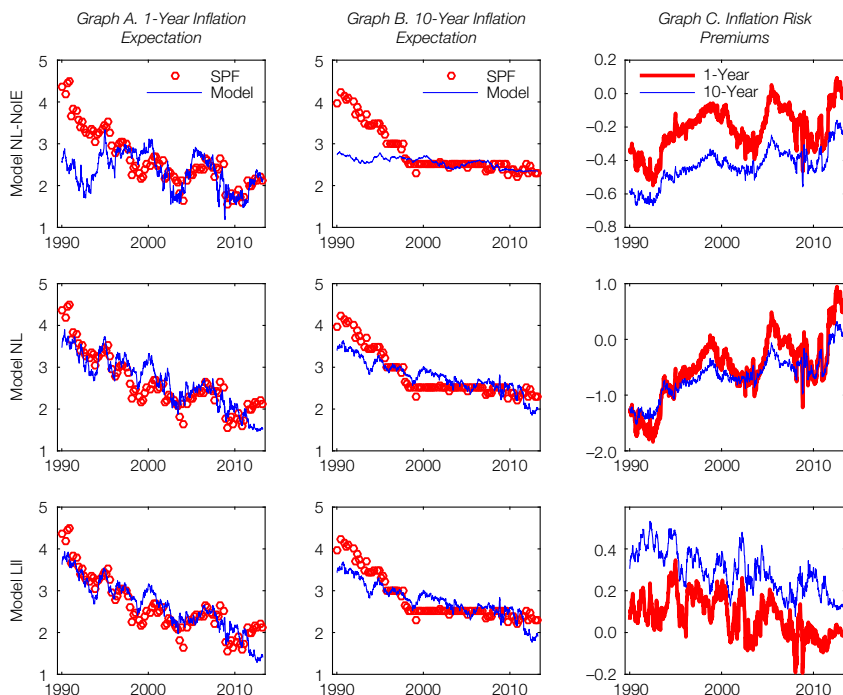
A visual comparison between the model-implied inflation expectations and survey forecasts, plotted in the top panels of Graphs A and B in Figure 5, show that Model NL-NoIE fails to capture the downward trend in survey inflation forecasts during much of the sample period and implies that the 10-year inflation expectation barely moved. This is the flip side of the discussions in Section V.B, where Model NL-NoIE implied a 10-year real rate that was too variable and explained the entire decline in nominal yields during the 1990s. Adding information from survey inflation forecasts brought the Model NL-implied inflation expectations more in line with survey forecasts, as can be seen from the middle panels of Graphs A and B. Model NL then explains the smaller decline in nominal yields by generating a sustained increase in inflation risk premiums from around -1.5% in the early 1990s to the current level of slightly above 0, in contrast to previous findings of overall positive and generally declining inflation risk premiums over the past 3 decades (see Section VII). By comparison, Model

²³For a brief account of the episode, see Hu and Worah (2009).

²⁴The R^2 is defined as the percentage of in-sample variations of each data series explained by the model. Unlike in a regression setting, a negative value of R^2 could arise because the model expectation and the prediction errors are not guaranteed to be orthogonal in a small sample.

FIGURE 5
Model-Implied Inflation Expectation and Inflation Risk Premiums

The three rows of Figure 5 plot results from Model NL-NoIE, Model NL, and Model LII, respectively. Graphs A and B plot the 1- and 10-year model-implied inflation expectation together with the Survey of Professional Forecasters (SPF) counterpart, respectively. Graph C plots the 1- and 10-year model-implied inflation risk premiums. All numbers are in percentage points.



LII, which allows for a TIPS-specific factor and is shown in the bottom panel of Graph C, generates 10-year inflation risk premiums that are mostly positive and fluctuate in the 0%–0.5% range and short-term inflation risk premiums that are fairly small and became persistently negative during the 2007–2008 financial crisis.

D. Out-of-Sample Forecasting

It is conceivable that a model with more parameters like Model LII could generate smaller in-sample fitting errors for variables whose fit is explicitly optimized but perform worse out of sample. To check this possibility, we run an out-of-sample forecasting horse race between the four term structure models estimated previously, a random-walk model, the monthly Michigan survey, the monthly Blue Chip Financial Forecasts (BCFF) survey, and the quarterly SPF, and we report the RMSEs in Table 5. Three conclusions emerge: First, the term structure models perform as well as the two professional surveys, SPF and BCFF, over a common sample period, and they perform much better than the Michigan survey and the random-walk model. Second, within the term-structure models, adding survey inflation forecasts helps improve the forecasting performance of

TABLE 5
Out-of-Sample Inflation Forecasting

Table 5 reports out-of-sample root mean squared errors (RMSEs), expressed in annualized percentage terms, from forecasting inflation over the next 1 and 2 years using five term structure models and a random-walk (RW) model, and over the next year from three surveys. The forecasting period is from Jan. 5, 2005, to Mar. 27, 2013. The term-structure model forecasts are constructed based on parameters estimated using all up-to-date data. The RW model assumes that inflation over the next 1 and 2 years will be the same as inflation over the previous 1 and 2 years, respectively. We also compare the models to the monthly Michigan (MICH) and Blue Chip Financial Forecasts (BCFF) surveys and the quarterly Survey of Professional Forecasters (SPF) survey on dates when those surveys are available. In particular, the monthly Michigan and BCFF surveys are assumed to be conducted on the third Wednesday of each month, and the quarterly SPF survey is assumed to be conducted on the third Wednesday of the middle month of each quarter.

		Models					Surveys		
Frequency	No. of Obs.	NL-NoIE	NL	LI	LII	Random Walk	MICH	BCFF	SPF
<i>Panel A. 1-Year Horizon</i>									
Weekly	430	1.4827	1.4936	1.5052	1.4689	2.3724	2.1778	1.5798	1.5144
Monthly	99	1.4787	1.4944	1.5062	1.4687	2.3691			
Quarterly	33	1.4369	1.4595	1.4857	1.4566	2.3020			
<i>Panel B. 2-Year Horizon</i>									
Weekly	430	0.9414	0.9030	0.8956	0.8849	1.4685			
Monthly	99	0.9416	0.9084	0.8995	0.8887	1.4683			
Quarterly	33	0.9129	0.8680	0.8600	0.8552	1.4059			

Model NL. Finally, Model LII outperforms Model NL, with the improvement more pronounced at the longer 2-year horizon.

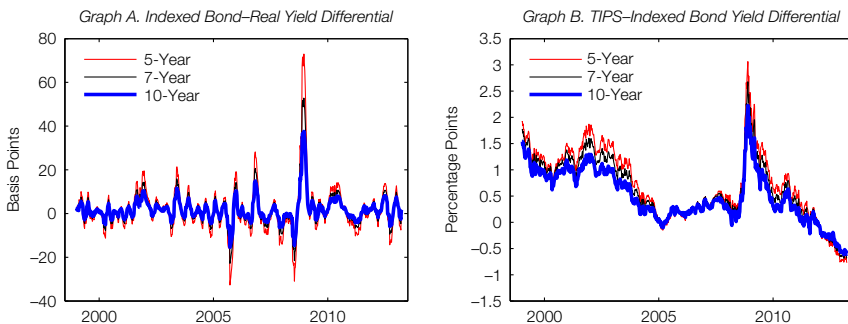
In addition, the robustness checks reported in the Internet Appendix show that the parameter estimates and good performance of Model LII remain largely intact when reestimated over a precrisis sample. Therefore, we mainly focus on this model in the remainder of our analysis.

E. The Effect of Indexation Lags

Graph A of Figure 6 shows the Model LII–implied differences between the indexed bond yields $y_{i,\tau,t}^I$, specified in equation (29), and the real yields $y_{i,\tau}^R$, specified in equation (15), for maturities of 5, 7, and 10 years. These estimates are generally small but rose to more than 30 bps at the 10-year maturity and 70 bps

FIGURE 6
Model-Implied Yield Spreads

Graphs A and B of Figure 6 plot Model LII–implied differences between indexed bond yields and real yields and between TIPS yields and indexed bond yields, respectively, for maturities of 5, 7, and 10 years.



at the 5-year maturity in Dec. 2008, when 3-month CPI inflation was running at an annual rate of about -13% . The differences between indexed bond yields and fully indexed real yields are estimated to be almost entirely due to the expected divergence between inflation over the 2.5 months prior to time t and inflation over the 2.5 months before the maturity of the bond, the second term on the left-hand side of equation (28). The last term, the indexation-lag premium, is estimated to be generally small, varying between -5 and 3 bps at the 10-year maturity, consistent with Risa (2001), and slightly larger at shorter maturities. Similar patterns are observed in all models we estimate.

VI. What Drives the TIPS Spread

In this section, we examine estimates of the TIPS spread from Model LII and show them to be mostly explained by proxies of TIPS liquidity. Other factors, such as CPI seasonality, TIPS deflation floors, and special demand for nominal Treasuries, also account for a small portion of their variations, although the effects of those factors appear to be much less significant.

A. Model Estimates of the TIPS Spread

The TIPS spread implied by Model LII is plotted in Graph B of Figure 6 for maturities of 5, 7, and 10 years. Three main features emerge: First, this spread exhibits substantial time variations at all maturities. Such variability at maturities as long as 10 years is in part due to the estimated high persistence of the TIPS-specific factor under the risk-neutral measure.²⁵ Second, the term structure of the estimated TIPS spread is downward sloping in 2001–2004 and 2008–2011. A market price of risk on the independent TIPS-specific factor that is on average positive, as is the case here, would contribute to a downward-sloping term structure of the TIPS spread.

Finally, the TIPS spreads were fairly high (1.5% – 2% range) when TIPS were first introduced but had been steadily declining until around 2004, likely reflecting the maturing process of a relatively new financial instrument. The spread surged to record-high levels ($\sim 3\%$) after the Lehman bankruptcy, reflecting a sharp increase in transaction and funding costs for TIPS as well as heightened risk aversion.²⁶ The heavy flight-to-safety flows into the nominal Treasury market likely also contributed to larger demand imbalances between nominal Treasuries and TIPS, which in turn should lead to a larger liquidity premium in TIPS yields relative to their nominal counterparts. Similarly sharp rises in liquidity premiums and/or illiquidity measures were seen in other key markets during the 2007–2008 financial crisis, including equity and corporate bond markets (see, e.g., Bao et al. (2011), Dick-Nielsen et al. (2012), and Brennan, Huh, and Subrahmanyam (2013)).

²⁵ As can be seen from Table 3, its risk-neutral persistence, $\tilde{\kappa}^* = \tilde{\kappa} + \tilde{\sigma} \tilde{\lambda}_1$, is estimated to be very small at around 0.1 in all models and is tightly estimated, with a standard error of about 0.006.

²⁶ These estimates are somewhat larger than those of Pflueger and Viceira (2016) but are broadly in line with those of AACMY (2016).

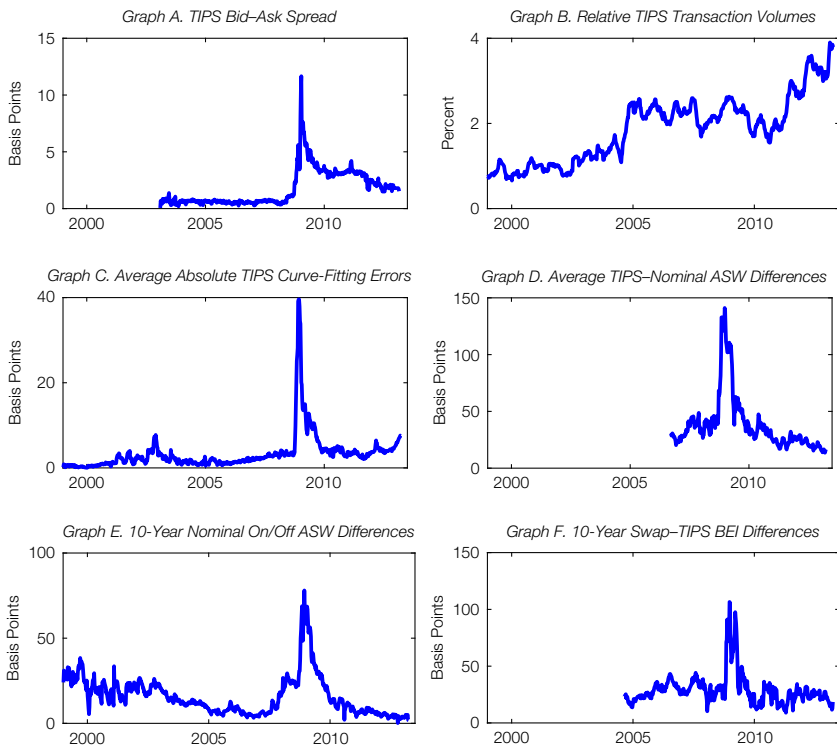
B. Link to Observable TIPS Liquidity Measures

Given the unobserved nature of the TIPS-specific factor in our model, one may question whether it is indeed capturing TIPS liquidity rather than other components of TIPS yields that are orthogonal to nominal Treasury yields. In this section, we provide strong evidence in favor of a liquidity premium interpretation of the TIPS spreads by linking them to various observable measures of TIPS liquidity while controlling for other factors that might have contributed to a wedge between TIPS yields and indexed bond yields.

One immediate difficulty we face is the lack of real-time, forward-looking measures of liquidity conditions in the TIPS market that are available over a reasonably long sample period. For example, as shown in Graph A of Figure 7, one widely used measure of illiquidity, the bid–ask spread, only became available for TIPS in 2003 from Tradeweb.²⁷ A measure that is available over a longer sample

FIGURE 7
Observable Measures of TIPS Liquidity

Figure 7 plots various measures of liquidity conditions in the TIPS market, including the 10-year TIPS bid–ask spread (Graph A), the TIPS trading volumes relative to those in nominal Treasury coupon securities (Graph B), the average mean-fitting errors from the Svensson TIPS yield curve (Graph C), the difference between the off-the-run and the on-the-run 10-year nominal Treasury par asset swap (ASW) spreads (Graph D), the average difference between TIPS and nominal Treasury ASW spreads (Graph E), and the difference between the 10-year inflation swap rate and the 10-year break-even inflation (BEI) (Graph F).



²⁷Tradeweb data, Thomson Reuters.

is the relative trading volumes of TIPS versus nominal Treasury coupon securities, plotted in Graph B.²⁸ This measure remained low up to mid-2004 and then rose substantially, suggesting steady improvement in TIPS liquidity during the precrisis sample period. The rise in relative trading volumes coincides roughly with the decline in our estimated TIPS spread over the same time window, with the two series showing a highly negative correlation of about -80% over the period 1999–2007.

Another observable measure of TIPS liquidity used in the literature is the average absolute fitting errors from the Svensson TIPS yield curve, plotted in Graph C of Figure 7. This measure captures funding constraints and limits to arbitrage that prevent investors from eliminating deviations of yields from their fundamental values as measured from a fitted yield curve.²⁹ It is plausible that during a financial crisis, capital becomes more scarce, and risk aversion runs higher, leaving significant arbitrage opportunities unexploited. According to this measure, liquidity conditions in the TIPS market were fine until the inception of the 2007–2008 financial crisis, when they suddenly deteriorated.

Three other considerations also affected our selection of liquidity proxies. First, as noted by Goldreich, Hanke, and Nath (2005), to capture liquidity premiums, we need measures not only of current liquidity conditions but, more importantly, of liquidity conditions expected to prevail over the life of the instruments. Second, unlike the bid–ask spread and the TIPS curve-fitting errors, the liquidity proxies relevant to our study should capture not the absolute level of TIPS (il)liquidity but, rather, their liquidity relative to nominal Treasury securities. Finally, the ideal proxies should also reflect variations in investors' risk attitude toward any given liquidity risk. To capture those additional dimensions of liquidity premiums, we examine two closely related measures based on asset prices. The first such measure is the difference between TIPS and off-the-run nominal asset swap (ASW) spreads, obtained from Barclays and plotted in Graph D of Figure 7.³⁰ Because both nominal Treasury securities and TIPS are usually considered free of default risk, their ASW spreads can be regarded as a good market-based measure of the liquidity premiums in those assets, and the difference between TIPS and nominal Treasury ASW spreads would be an ideal measure of the relative illiquidity of TIPS.³¹ Unfortunately, TIPS ASWs only started trading in 2006; we therefore use the difference between the off-the-run and on-the-run 10-year nominal Treasury ASW spreads, obtained from JP Morgan and shown in Graph E, as an approximation when studying the full sample. The correlation between the two measures of ASW spread differences is 0.93 since 2006 when both

²⁸We construct the measure as 13-week averages of weekly dealer transaction volumes in TIPS divided by those in nominal Treasury coupon securities using data reported by primary dealers and collected by the Federal Reserve Bank of New York under Government Securities Dealers Reports (FR-2004).

²⁹Similar measures have been used by Fontaine and Garcia (2012) and Hu, Pan, and Wang (2013) to measure the liquidity of nominal Treasury securities, and by GH (2013) for TIPS.

³⁰In a fixed-income asset swap, one party exchanges the fixed-rate cash flows from the underlying security for a floating-rate cash flow, where the floating rate is typically quoted as 6-month London Interbank Offered Rate (LIBOR) plus a spread, the asset swap spread.

³¹Such a measure is used in a recent study by Campbell, Shiller, and Viceira (2009), which focuses on a more recent sample of July 2007–Apr. 2009.

are available. As can be seen from Graphs D and E, both measures spiked during the crisis, reflecting general funding pressures leading to a dramatic divergence between prices of securities with only small liquidity differentials. The second forward-looking measure of TIPS liquidity premiums is the difference between inflation swaps and TIPS BEI, available since late 2004 and plotted in Graph F.³² As noted by Campbell et al. (2009), in theory this measure is linked to the difference between the TIPS and nominal ASW spreads through a no-arbitrage relationship; empirically, the two measures have a high correlation of 88% in the post-2006 sample when both are available, although the swap–BEI difference exhibits a smaller spike during the 2007–2008 crisis and also an earlier return to its normal levels afterward.

To quantify the effects of these factors, we run univariate and multivariate regressions of the 10-year TIPS spread from Model LII on various liquidity measures. Panel A of Table 6 examines the three measures that are available over the full sample period: the relative TIPS-nominal trading volumes, the nominal on- and off-the-run ASW spread difference, and the average TIPS curve-fitting errors. The coefficients on the three variables all carry the expected signs and are statistically significant. In particular, one can expect TIPS liquidity premiums to be lower when TIPS transaction volumes rise relative to those of nominal Treasury securities, when nominal on- and off-the-run ASW spreads trade closer to each other, and/or when TIPS yields show smaller deviations from their fundamental values. Together these three variables explain 85% of the variation in the 10-year TIPS spread estimates.

Panel B of Table 6 expands the regressions to include all six liquidity measures over the sample period since Sept. 20, 2006, when all measures became available. Again, most coefficients are of the expected sign and statistically significant, except for the TIPS bid–ask spread and the swap–BEI difference, which are significant on their own but become insignificant when all TIPS liquidity measures are included. The magnitude of the coefficients on the relative TIPS trading volumes is smaller in this more recent sample period, consistent with the intuition that the “growing pains” of the TIPS market is a more important story in the earlier part of the sample. In the univariate regression, the coefficient on the TIPS fitting errors nearly doubled, and the regression now explains a much larger portion of TIPS spread variations, arguably reflecting the importance of funding constraints and limits to arbitrage during the 2007–2008 financial crisis. The same three variables as in Panel A explain a very similar percentage (83%) of the TIPS spread variations, with this number rising to 86.3% when all observable liquidity measures (except the nominal on- and off-the-run ASW spread difference) are included.

The results from Table 6 confirm that the model-implied TIPS spreads are indeed capturing current and expected future relative liquidity conditions in the

³²Inflation swaps are over-the-counter swap contracts under which one party pays a fixed rate, the inflation swap rate, and the other party pays a floating rate that equals the realized inflation rate. Inflation swap rates are usually above the TIPS BEI at the same maturity due to liquidity differences between the TIPS and nominal Treasury markets as well as the lack of liquidity in the inflation swaps market. Fleckenstein et al. (2014) and Christensen and Gillan (2012) examine these explanations in more detail.

TABLE 6
What Drives the TIPS Spreads

Panel A of Table 6 reports results from univariate and multivariate regressions of the model-implied 10-year TIPS spread on the relative TIPS trading volume, the nominal on/off-the-run asset swap (ASW) spread difference, and the average TIPS curve-fitting error using weekly data from Jan. 6, 1999, to Mar. 27, 2013. Panel B reports results from univariate and multivariate regressions of the model-implied 10-year TIPS spread on all TIPS liquidity measures from Sept. 20, 2006, to Mar. 27, 2013. Finally, the last column adds to the joint regression other potential drivers of the TIPS break-even inflation (BEI), including seasonality and deflation floor adjustment to 10-year TIPS yield, the one-month-ahead option-implied volatility on the S&P 500 index (VIX), and the difference in repo rates on 10-year on-the-run nominal Treasury notes and TIPS. Ordinary least squares (OLS) standard errors are reported in parentheses. * and ** indicate significance at the 5% and 1% levels, respectively.

Panel A. Regression Analysis: Full Sample

	1	2	3	4
Constant	1.3550** (0.0342)	-0.0191 (0.0197)	0.4022** (0.0220)	0.9109** (0.0327)
Relative TIPS transaction volume	-0.4578** (0.0170)			-0.4215** (0.0136)
Nominal on/off ASW spread difference		0.0332** (0.0010)		0.0183** (0.0011)
Average TIPS curve-fitting error			0.0309** (0.0040)	0.0264** (0.0028)
No. of obs.	743	743	743	743
Adj. R^2	49.4%	59.5%	7.4%	84.8%

Panel B. Regression Analysis: Since 2006

	1	2	3	4	5	6	7	8	9
Constant	-0.0765 (0.0419)	1.3680** (0.1161)	-0.0334 (0.0277)	-0.1410** (0.0214)	-0.4097** (0.0238)	-0.4112** (0.0423)	0.7900** (0.0642)	0.4153** (0.0604)	0.3359** (0.0651)
10-year TIPS bid-ask spread	0.1601** (0.0143)							0.0030 (0.0074)	0.0106 (0.0085)
Relative TIPS transaction volume		-0.4402** (0.0467)					-0.3751** (0.0249)	-0.3014** (0.0224)	-0.2916** (0.0239)
Average TIPS curve-fitting error			0.0620** (0.0035)				0.0291** (0.0038)	0.0086* (0.0035)	0.0090* (0.0037)
Nominal on/off ASW spread difference				0.0290** (0.0010)			0.0173** (0.0015)		
TIPS-nominal ASW spread difference					0.0187** (0.0005)			0.0177** (0.0013)	0.0180** (0.0017)
Inflation swaps-BEI difference						0.0241** (0.0013)		-0.0036* (0.0015)	-0.0045** (0.0017)
Deflation floor adjustment									0.0045 (0.0039)
Seasonality adjustment									0.0057* (0.0028)
VIX									0.0016 (0.0016)
Nominal-TIPS repo spread									0.0007** (0.0002)
TIPS purchase dummy									-0.0118 (0.0309)
No. of obs.	337	341	341	341	341	341	341	337	326
Adj. R^2	27.0%	20.5%	47.6%	71.1%	78.8%	51.2%	82.7%	86.3%	88.6%

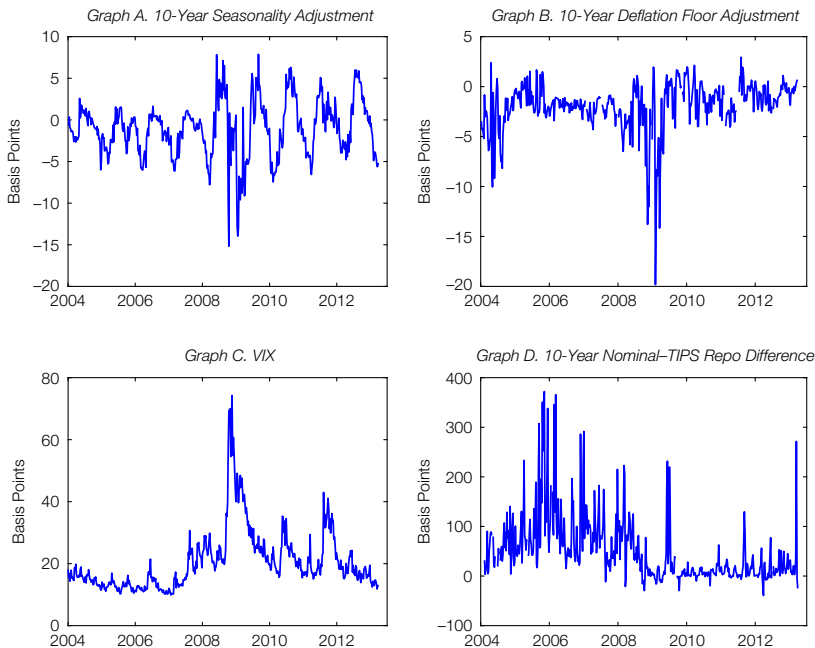
TIPS market as well as the associated risk premiums. We caution, though, that the linear regression analysis may not capture the more complex relation that quantities like bid-ask spreads and trading volumes can be expected to have with liquidity premiums. In comparison, the latent-liquidity-factor approach used in this article has the advantage of being more flexible, without the need to assume a rigid link between the liquidity spreads and one or more observable measures. That said, our regression analysis suggests that the difference between the TIPS and nominal ASW spreads stands out as the most promising real-time, observable measure of TIPS liquidity premiums, at least based on data since 2006.

C. The Importance of Other Drivers

In addition to lower liquidity and the indexation lag, a few other aspects of TIPS might also affect their pricing relative to nominal Treasury securities. First, TIPS are indexed to non-seasonally-adjusted CPI; therefore, TIPS with a base reference month typically associated with higher (lower) inflation than the current reference month can be expected to trade at a slightly higher (lower) price, all else equal. To capture this effect, we estimate a seasonally adjusted TIPS yield curve and compute the difference between the actual 10-year TIPS yield and the seasonally adjusted 10-year TIPS yield, plotted in Graph A of Figure 8.³³ This measure fluctuates between plus and minus 10 bps and tends to be positive in the summer months and negative during the winter months, which, after taking the indexation lag into account, is consistent with the pattern in recent years that CPI tends to be higher from March to May and lower from October to December.

FIGURE 8
Other Potential Drivers of TIPS BEI

Figure 8 plots measures of other factors that might affect the TIPS break-even inflation (BEI), including the incremental change in the 10-year TIPS yield after seasonality adjustment (Graph A) and adjustment for the deflation floor (Graph B), the probability of a flight-to-safety episode (Graph C), and the difference between the repo rates on 10-year on-the-run nominal Treasury notes and TIPS (Graph D).



³³We adjust the quoted real prices and future real coupon and principal payments by the expected seasonal factor, which is proxied by the average ratio of non-seasonally-adjusted CPI to seasonally adjusted CPI over the previous 5 years. A detailed description of the methodology can be found in Grishchenko, Li, and Vollmer (2015).

Second, TIPS contain an embedded put option, often called the “deflation floor,” because the principal repayment cannot fall below par.³⁴ This option is far out of the money and of little value in normal times, but it could move into the money when deflation becomes a concern, especially for newly issued TIPS with reference CPIs close to the current level.³⁵ At the height of the 2007–2008 financial crisis, valuable deflation protection would push up TIPS prices and push down TIPS yields, partially offsetting the effect of higher TIPS liquidity premiums. The TIPS yields used in this study are based on a yield curve fitted to the entire universe of TIPS and are therefore less susceptible to the effect of the deflation floor than on-the-run TIPS yields. Nevertheless, to examine the remaining effect of the embedded deflation floor, we calculate the difference between the 10-year TIPS yield from the seasonally adjusted curve fitted to all TIPS and the 10-year yield from a seasonally adjusted curve estimated without the on-the-run and first-off-the-run 5- and 10-year TIPS. This difference, shown in Graph B of Figure 8, was less than 5 bps in magnitude in normal times but widened to –20 bps during the 2007–2008 financial crisis.³⁶

Finally, any special demand for nominal Treasury securities would push down nominal Treasury yields as well as any estimates of real yields that are consistent with those nominal yields, leading to a wider gap between TIPS yields and model-implied real yields. This is most obvious during times of market panics with large flight-to-safety flows into the nominal Treasury market but can also arise in normal times due to the special role nominal Treasury securities play in facilitating hedging and short covering activities and in satisfying regulatory requirements. Such special demand is closely linked to the relative liquidity between nominal Treasuries and TIPS but might merit a separate examination. To capture the safe-haven flows into nominal Treasuries, we use the one-month-ahead option-implied volatility on the S&P 500 index (VIX) plotted in Graph C of Figure 8. To capture the more general concept of convenience yield in nominal Treasuries, we look at the difference between the overnight repo rates for the on-the-run 10-year nominal Treasury and the on-the-run 10-year TIPS, plotted in Graph D.³⁷ Motivated by Christensen and Gillan (2017), we also include a time dummy that equals 1 in the weeks during which the Federal Reserve purchased TIPS as part of its various asset purchase programs during the recent financial crisis, and 0 otherwise.

³⁴See Jacoby and Shiller (2008), Wright (2009), Christensen et al. (2010), and Grishchenko, Vanden, and Zhang (2016) for more details.

³⁵For example, Wright (2009) showed that two TIPS with similar maturity dates but different issue dates, the Apr. 2013 TIPS issued in 2008 and the July 2013 TIPS issued in 2003, were traded at nearly identical real yields prior to Sept. 2008, but the yield spread widened to as much as 200 bps in Dec. 2008 as the price of the more recently issued Apr. 2013 TIPS rose, reflecting the higher value of the deflation floor, while the price of the July 2013 TIPS dropped.

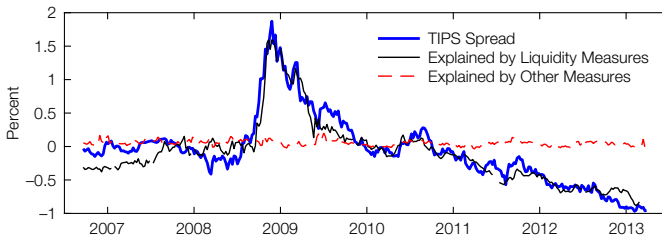
³⁶Alternatively, one could derive the value of the embedded deflation floor in any individual TIPS using the risk-adjusted PDF of inflation constructed from inflation caps and floors, as in Kitsul and Wright (2013) and Fleckenstein, Longstaff, and Lustig (FLL) (2017). However, quotes on inflation caps and floors have been available only since 2009. In addition, the effect of the embedded deflation floor on individual TIPS yields is likely to differ from the effect on fitted TIPS yields due to yield curve-fitting errors.

³⁷Convenience yields on nominal Treasury securities are studied by Longstaff (2004), Krishnamurthy and Vissing-Jorgensen (2012), and Bansal, Coleman, and Lundblad (2011), among others.

The last column in Panel B of Table 6 adds those five measures to the joint regression. They all carry the expected signs but, with the exception of the repo difference, are not statistically significant at the 1% level and together add limited additional explanatory power for the TIPS spread. Indeed, as can be seen from Figure 9, the regression attributes almost all the variation in the TIPS spread to TIPS liquidity measures, whereas the contribution from the five other measures is essentially flat and close to 0. This does not imply that risk aversion or the relative demand imbalances have no effect on the TIPS spread but suggests those effects are largely subsumed in the price-based liquidity measures. Overall, these results confirm our conjecture that the TIPS spread is overwhelmingly reflecting a liquidity premium in TIPS yields. In the remainder of the article, we use the terms “TIPS spread” and “TIPS liquidity premiums” interchangeably.

FIGURE 9
Contributions from TIPS Liquidity and Other Measures

Figure 9 shows the relative contributions from the liquidity measures and other measures based on regression 9 in Panel B of Table 6. The thick solid line plots the Model LII-implied TIPS spread minus the regression constant, and the thin solid line and the dotted line plot the portions explained by TIPS liquidity measures and other measures, respectively.



D. Economic Significance of TIPS Liquidity Premiums

We assess the economic significance of the TIPS liquidity premiums by examining the proportions of variations in TIPS yields and TIPS BEI that can be attributed to variations in those premiums. Using equations (1), (27), and (D-7), we can decompose TIPS yields, $y_{t,\tau,d}^T$, and TIPS BEI, $BEI_{t,\tau,d}^T$, into different components:

$$(31) \quad \begin{aligned} y_{t,\tau,d}^T &= y_{t,\tau}^R + (y_{t,\tau,d}^I - y_{t,\tau}^R) + L_{t,\tau,d}, \\ BEI_{t,\tau,d}^T &= I_{t,\tau} + \wp_{t,\tau} - (y_{t,\tau,d}^I - y_{t,\tau}^R) - L_{t,\tau,d}, \end{aligned}$$

where $y_{t,\tau}^R$ is the underlying real yield, $y_{t,\tau,d}^I - y_{t,\tau}^R$ represents the effect of the indexation lag, $L_{t,\tau,d}$ is the TIPS liquidity premium, $I_{t,\tau}$ is expected inflation over the next τ periods, and $\wp_{t,\tau}$ is the inflation risk premium. Table 7 reports an in-sample variance decomposition based on equation (31) and Model LII estimates. A time-series plot of the decomposition is shown in Figure 10.

Real yields dominate TIPS liquidity premiums in accounting for the time variations in TIPS yields. By contrast, the TIPS liquidity premium is a more important driver of TIPS BEI, explaining 36%–49% of its movement across the three maturities, although expected inflation and inflation risk premiums jointly still

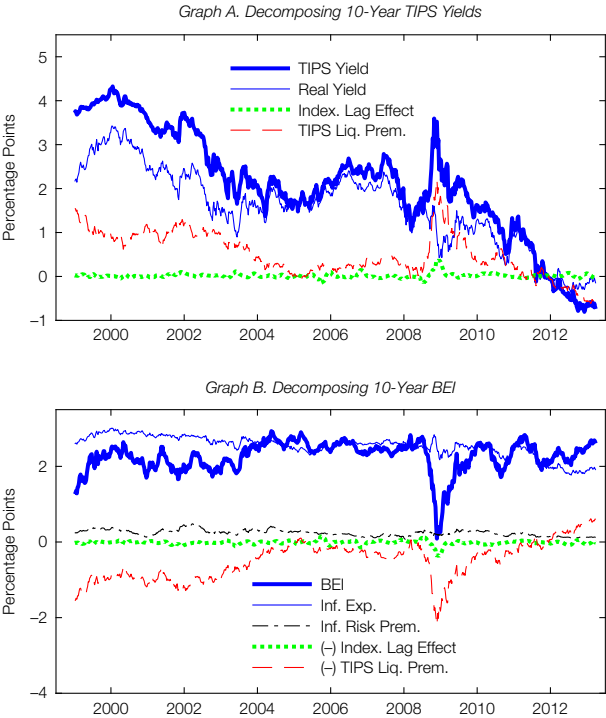
TABLE 7
In-Sample Variance Decomposition of TIPS Yields and TIPS BEI

Table 7 reports the in-sample variance decompositions of TIPS yields into real yields, indexation-lag effects, and TIPS liquidity premiums and of nominal yields into expected inflation, inflation risk premiums, indexation-lag effects, and the negative of TIPS liquidity premiums, all based on Model LII estimates and expressed in percentage points. The variance decompositions are calculated as the in-sample covariance between TIPS yields (break-even inflation (BEI)) and the individual components, divided by the in-sample variance of TIPS yields (BEI). Standard errors calculated using the delta method are reported in parentheses.

Maturity	TIPS Yield			TIPS BEI			
	Real Yield	Indexation Lag	Liquidity Premium	Expected Inflation	Inflation Risk Premium	Indexation Lag	Liquidity Premium
5-year	83.1 (5.5)	0.6 (1.2)	16.3 (5.4)	38.5 (2.4)	7.0 (6.8)	5.9 (3.3)	48.6 (6.3)
7-year	83.2 (5.6)	0.3 (1.2)	16.5 (5.5)	44.5 (2.8)	8.5 (9.0)	4.6 (2.8)	42.4 (7.6)
10-year	84.0 (5.7)	0.1 (1.2)	15.9 (5.6)	49.6 (5.2)	11.1 (11.8)	3.5 (2.5)	35.9 (9.4)

FIGURE 10
Decomposing TIPS Yields and TIPS BEI

Graph A of Figure 10 decomposes the 10-year TIPS yield into the real yield and the TIPS liquidity premium, and Graph B decomposes the 10-year TIPS break-even inflation (BEI) into expected inflation, the inflation risk premium, and the TIPS liquidity premium, both according to equation (31) and Model LII estimates.



account for the majority of time variations in TIPS BEI at 7- and 10-year maturities. Our results suggest that one should be especially cautious in interpreting fluctuations in TIPS BEI solely in terms of changes in inflation expectations or inflation risk premiums.

VII. Inflation Risk Premiums and Expected Inflation: Comparison across Studies

In this section, we compare our estimates of inflation risk premiums and expected inflation with those obtained in previous and concurrent studies, collected in Table 8. We organize those studies based on the length of the sample periods, ordering last those that focus on more recent samples. For example, we start with the estimates of ABW (2008), which cover the period from 1952 to 2004, and conclude with those of FLL (2017), which focus on the recent decade of 2004–2014. This organizing principle is motivated by Bekaert and Wang (2010), who find that the longer the sample period, the larger and more robustly positive are the estimates of the inflation risk premium.

To understand the wide range in the estimates of the inflation risk premium in Table 8, it is helpful to start from Kitsul and Wright ((2013), Figs. 12 and 13), who observe that the empirical pricing kernel appears to be U-shaped in inflation. This suggests that when the economy is close to deflation (hyperinflation), a pickup in inflation and the resulting losses to nominal bondholders are likely to coincide with lower (higher) marginal utility of wealth, as reflected in the downward-sloping (upward-sloping) arm of the U. As a result, inflation risk

TABLE 8
Estimates of Inflation Risk Premiums and Expected Inflation

Table 8 reports estimates from ABW (2008), BJ (2005), CM (2012), HPR (2012), ABC (2012), this article (DKW, in bold), AACMY (2016), GH (2013), and FLL (2017). HPR estimates are obtained from the Web site of the Federal Reserve Bank of Cleveland (<https://www.clevelandfed.org/our-research/indicators-and-data/inflation-expectations.aspx>). GH generate a range of estimates depending on the proxy used for expected inflation; the table reports the results using past average 5- and 10-year realized inflation as proxies for future 5- and 10-year expected inflation.

	Level of Inflation in the Estimation Sample								
	High			High and Low			Low		
	ABW	BJ	CM	HPR	ABC	DKW	AACMY	GH	FLL
<i>Panel A. Inflation Risk Premiums (basis points)</i>									
5-year	114	42	NA	17	16	19	−17	−36	−4.5
10-year	NA	70	67	45	24	29	10	−12	2.8
<i>Panel B. Expected Inflation (%)</i>									
5-year	3.94	NA	NA	3.13	2.88	2.73	2.56	2.51	2.36
10-year	NA	NA	4.05	3.10	2.88	2.77	2.78	2.55	2.58
<i>Panel C. Data Used</i>									
TIPS			Yes			Yes	Yes	Yes	
Inflation swaps				Yes					Yes
Survey inflation forecasts			Yes	Yes		Yes			
<i>Panel D. Sample Period</i>									
Start year	1952	1960	1971	1982	1985	1990	1999	2000	2004
End year	2004	2000	2008	2010	2011	2013	2013	2008	2014

premiums are likely to be lower or even negative during times of deflation risks and higher and positive when hyperinflation is a concern.³⁸

Historically, the 1970s and early 1980s saw countercyclical, high, and volatile inflation as well as unstable inflation expectations, which tend to be associated with hyperinflation risks and high inflation risk premiums. In addition, survey information shows that inflation uncertainty was elevated up to the early 1990s (D'Amico and Orphanides (2014)), likely further boosting inflation risk premiums. Subsequently, inflation expectations declined and stabilized at much lower levels after 1998, as evidenced in the SPF and other surveys, together with realized inflation, which can be expected to lead to a gradual decline in inflation risk premiums over this period. Finally, since the last crisis, realized inflation has been running persistently low, and both surveys and inflation derivatives attached non-negligible odds to near- or below-zero inflation at horizons even above 5 years, which may result in very low or even negative inflation risk premiums.

It is perhaps not surprising, then, that the studies reported in Table 8 using data from the 1970s and early 1980s but not from the 2007–2008 financial crisis, including those of ABW (2008), BJ (2005), and CM (2012), obtain estimates of inflation risk premiums that are mostly positive and large in magnitude. In contrast, models estimated over shorter sample periods that do include the financial crisis, including AACMY (2016), GH (2013), and FLL (2017), produce lower estimates of inflation risk premiums that are often negative at shorter maturities. The evidence shown in our article also suggests that any studies using TIPS but ignoring the liquidity mismatch between nominal Treasuries and TIPS are likely to underestimate the inflation risk premiums.³⁹

Finally, studies like those by HPR (2012) and Ajello, Benzon, and Chyruk (ABC) (2012) and ours (DKW) that include both the period of the 1980s and/or early 1990s, when inflation gradually comes down and stabilizes, and the latest financial crisis produce mostly positive estimates of inflation risk premiums, especially at longer maturities, that lie between those of the two groups of studies mentioned previously. HPR obtained estimates for maturities up to 29 months that frequently turned negative during episodes of financial market turmoil, perhaps reflecting, as they note, the relative attractiveness of safe and liquid shorter-maturity Treasuries. This again suggests that the special features of nominal Treasuries, and the lack thereof in TIPS or inflation swaps, should be taken into account when estimating the inflation risk premiums.

Overall, our estimated inflation risk premiums fall in the middle of a rather wide range of estimates found in other studies. Similar to the broad pattern of lower values in more recent samples, our estimates of the inflation risk premium trended down over time and briefly turned negative at the 5-year maturity at the height of the 2007–2008 financial crisis, likely reflecting the substantial risk of deflation at the time.

³⁸ Similar points are made by Campbell et al. (2017) and David and Veronesi (2013).

³⁹ Indeed, GH (2013) report negative inflation risk premiums even at longer maturities when TIPS are taken at their face value. Once a regression-based TIPS liquidity adjustment is made, the estimated 10-year inflation risk premium turns slightly positive.

Similarly, the average inflation expectation estimates reported in Table 8 also depend strongly on the sample period: Studies using a longer sample that spans the 1970s and 1980s but ends before the 2007–2008 financial crisis, those of ABW (2008) and CM (2012), obtain the highest inflation expectation estimates. Studies that focus on the post-1998 period of low and stable actual and expected inflation, including those by AACMY (2016), GH (2013), and FLL (2017), obtain estimates that are much smaller in magnitude. Finally, studies that include both the early 1990s and the latest financial crisis (HPR (2012), ABC (2012)) report estimates that are roughly in line with our study (see also Fig. 6, Panel B, in HPR (2012)).

VIII. Conclusion

This article shows that a TIPS-specific factor is important for explaining TIPS yield and TIPS BEI variations but not for nominal yield changes, and it provides evidence that this factor is linked to the relative illiquidity of TIPS versus nominal Treasury securities.

In particular, we develop a joint no-arbitrage term structure model of nominal yields, TIPS yields, and realized inflation and show that ignoring the spread between TIPS and risk-free real yields leads to a poor model fit of TIPS yields, TIPS BEI, and survey inflation forecasts. In models allowing such a spread, its estimated values were fairly large ($\sim 1\%$) until about 2003 and fluctuated within narrow ranges between then and the onset of the crisis, consistent with the common perception that TIPS market liquidity had steadily improved over time. The estimated TIPS spread shot up to nearly 300 bps after the Lehman collapse, amid reports of fire sales of TIPS and stringent funding liquidity conditions over that period. Consistent with these observations, regression analysis shows that about 85% of the variation in the estimated TIPS-indexed bond spread is explained by changes in observable measures of TIPS liquidity, whereas other factors, such as TIPS deflation floor and CPI seasonality, appear much less important.

TIPS BEI has increasingly gained attention as a measure of investors' inflation expectations that is available in real time and at high frequencies. However, our results raise caution in interpreting movements in TIPS BEI solely in terms of changing inflation expectations because substantial liquidity premiums and inflation risk premiums could drive a large wedge between the two, as demonstrated vividly during the 2007–2008 financial crisis. A better understanding of the determinants of TIPS liquidity premiums and the sources of its variation remains an interesting topic for future research.

Appendix A. Formulae and Notations

1. The Real Pricing Kernel

Applying Ito's lemma to equation (8) and using equations (3)–(7), the real pricing kernel can be derived as following the process

$$\begin{aligned}
 \text{(A-1)} \quad \frac{dM_t^R}{M_t^R} &= \frac{dM_t^N}{M_t^N} + \frac{dQ_t}{Q_t} + \frac{dM_t^N}{M_t^N} \times \frac{dQ_t}{Q_t} \\
 &= -r^R(x_t)dt - \lambda^R(x_t)'dB_t - (\cdot)dB_t^\perp,
 \end{aligned}$$

where the real short rate and the real prices of risk are given by

$$(A-2) \quad r^R(x_t) = \rho_0^R + \rho_1^R x_t,$$

$$(A-3) \quad \lambda^R(x_t) = \lambda_0^R + \Lambda^R x_t,$$

and the coefficients are linked to their nominal counterparts by

$$(A-4) \quad \rho_0^R = \rho_0^N - \rho_0^\pi - \frac{1}{2}(\sigma_q' \sigma_q + \sigma_q^{\perp 2}) + \lambda_0^{N'} \sigma_q,$$

$$(A-5) \quad \rho_1^R = \rho_1^N - \rho_1^\pi + \Lambda^{N'} \sigma_q,$$

$$(A-6) \quad \lambda_0^R = \lambda_0^N - \sigma_q, \quad \Lambda^R = \Lambda^N.$$

Appendix B. TIPS–Real Yield Spread

Because $\tilde{x}_t$ is independent of the other state variables in x_t , the first term on the right-hand side of equation (22) can be written as the sum of two components:

$$(B-1) \quad -\frac{1}{\tau} \ln E_t^Q(e^{-\int_t^{t+\tau} \tilde{\gamma} \tilde{x}_s ds}) - \frac{1}{\tau} \ln E_t^Q(e^{-\int_t^{t+\tau} (\rho_0^R + (\rho_1^R + \gamma)' x_s) ds}).$$

The first component can be solved to be

$$(B-2) \quad -\frac{1}{\tau} \ln E_t^Q(e^{-\int_t^{t+\tau} \tilde{\gamma} \tilde{x}_s ds}) = \tilde{a}_\tau + \tilde{b}_\tau \tilde{x}_t,$$

where $\tilde{a}$ and $\tilde{b}$ have the familiar form of factor loadings in a 1-factor Vasicek model:

$$(B-3) \quad \tilde{a}_\tau = \tilde{\gamma} \left[\left(\tilde{\mu}^* - \frac{\tilde{\sigma}^2}{2\tilde{\kappa}^*} \right) (1 - \tilde{b}_\tau) + \frac{\tilde{\sigma}^2}{4\tilde{\kappa}^*} \tau \tilde{b}_\tau^2 \right],$$

$$(B-4) \quad \tilde{b}_\tau = \tilde{\gamma} \frac{1 - \exp(-\tilde{\kappa}^* \tau)}{\tilde{\kappa}^* \tau},$$

in which $\tilde{\kappa}^* = \tilde{\kappa} + \tilde{\sigma} \tilde{\lambda}_1$ and $\tilde{\mu}^* = (\tilde{\kappa} \tilde{\mu} - \tilde{\sigma} \tilde{\lambda}_0) / \tilde{\kappa}^*$.

The second component can be shown to take the form

$$(B-5) \quad -(1/\tau) \ln E_t^Q(e^{-\int_t^{t+\tau} (\rho_0^R + (\rho_1^R + \gamma)' x_s) ds}) = a_\tau^T + b_\tau^{T'} x_t,$$

where $a_\tau^T = -A_\tau^T / \tau$ and $b_\tau^{T'} = -B_\tau^{T'} / \tau$ are given by

$$(B-6) \quad \frac{dA_\tau^T}{d\tau} = -\rho_0^R + B_\tau^{T'} (\mathcal{K} \mu - \Sigma \lambda_0^R) + \frac{1}{2} B_\tau^{T'} \Sigma \Sigma' B_\tau^{T'},$$

$$(B-7) \quad \frac{dB_\tau^{T'}}{d\tau} = -(\rho_1^R + \gamma) - (\mathcal{K} + \Sigma \Lambda^R)' B_\tau^{T'},$$

with initial conditions $A_0^T = 0$ and $B_0^{T'} = 0_{3 \times 1}$.

Taken together, we have

$$(B-8) \quad \begin{aligned} L_{t,\tau} &= (\tilde{a}_\tau + a_\tau^T) + [b_\tau^{T'} \quad \tilde{b}_\tau] \begin{bmatrix} x_t \\ \tilde{x}_t \end{bmatrix} - y_t^R \\ &= [\tilde{a}_\tau + (a_\tau^T - a_\tau^R)] + [(b_\tau^{T'} - b_\tau^R)' \quad \tilde{b}_\tau] \begin{bmatrix} x_t \\ \tilde{x}_t \end{bmatrix}, \end{aligned}$$

where the second equality comes from equation (15).

Appendix C. A 4-Factor Framework

The models with a TIPS-specific factor can be restated in a 4-factor Gaussian framework, in which the expanded state variables $z_t = [x'_t, \tilde{x}_t - \tilde{\mu}]$ follow the process

$$(C-1) \quad dz_t = \mathcal{K}_z(\mu_z - z_t)dt + \Sigma_z dB_t^z,$$

where

$$\mathcal{K}_z = \begin{bmatrix} \mathcal{K} & 0_{3 \times 1} \\ 0_{1 \times 3} & \tilde{\kappa} \end{bmatrix}, \quad \mu_z = \begin{bmatrix} \mu \\ 0 \end{bmatrix}, \quad \Sigma_z = \begin{bmatrix} \Sigma & 0_{3 \times 1} \\ 0_{1 \times 3} & \tilde{\sigma} \end{bmatrix}, \quad B_t^z = \begin{bmatrix} B_t \\ W_t \end{bmatrix}.$$

We de-mean $\tilde{x}_t$ so that the new 4-factor model continues to satisfy the identification condition $\mu_z = 0$. Parameters governing the real and nominal short rates and prices of risks are redefined in obvious ways. In particular, the real and nominal prices or risks are given by

$$(C-2) \quad \lambda_0^i = \begin{bmatrix} \lambda_0^i \\ \tilde{\lambda}_0 + \tilde{\lambda}_1 \tilde{\mu} \end{bmatrix}, \quad \Lambda^i = \begin{bmatrix} \Lambda^i & 0 \\ 0_{1 \times 3} & \tilde{\lambda}_1 \end{bmatrix}, \quad i = N, R.$$

TIPS can then be priced as real bonds similar to equations (12)–(14) based on the adjusted real short rate:

$$(C-3) \quad \begin{aligned} \tilde{r}_t^R &= r_t^R + l_t = \tilde{\rho}_0^R + \tilde{\rho}_1^{R'} z_t, \\ \tilde{\rho}_0^R &= \rho_0^R + \tilde{\gamma} \tilde{\mu}, \quad \tilde{\rho}_1^R = [\rho_1^R + \gamma \quad \tilde{\gamma}]. \end{aligned}$$

For reasons that will become clear in Appendix D, we also define the adjusted nominal short rate based on equations (A-4)–(A-5) as:

$$(C-4) \quad \begin{aligned} \tilde{r}_t^N &= r_t^N + l_t = \tilde{\rho}_0^N + \tilde{\rho}_1^{N'} z_t, \\ \tilde{\rho}_0^N &= \rho_0^N + \tilde{\gamma} \tilde{\mu}, \quad \tilde{\rho}_1^N = [\rho_1^N + \gamma \quad \tilde{\gamma}]. \end{aligned}$$

Appendix D. Indexed Bond Pricing

We adjust TIPS yields in equation (27) for the indexation lag in three steps. First, following Evans (1998) and Risa (2001) and as shown in equation (D-1), the artificial indexed bond can be viewed as a claim to a real payoff that equals the time $(T-l)$ price of a l -period nominal bond, $P_{T-l,l}^N$, to be received at time $T-l$:

$$(D-1) \quad \begin{aligned} \tilde{P}_{t,\tau,l}^I &= E_t \left[\frac{M_t^N}{M_t^N} \frac{Q_{T-l}}{Q_t} \right] = E_t \left[\frac{M_{T-l}^N}{M_t^N} E_{T-l} \left(\frac{M_T^N}{M_{T-l}^N} \right) \frac{Q_{T-l}}{Q_t} \right] \\ &= E_t \left[\frac{M_{T-l}^N}{M_t^N} \frac{Q_{T-l}}{Q_t} P_{T-l,l}^N \right] = E_t \left[\frac{M_{T-l}^R}{M_t^R} P_{T-l,l}^N \right]. \end{aligned}$$

By the same argument as in deriving the real bond price formula, we can show that the time t price of such an artificial indexed bond is given by

$$(D-2) \quad \tilde{P}_{t,\tau,l}^I = \exp(A_{\tau,l} + B'_{\tau,l} x_t),$$

where $A_{\tau,l}$ and $B_{\tau,l}$ satisfy the ordinary differential equation:

$$(D-3) \quad \frac{dA_{\tau,l}}{d\tau} = -\rho_0^R + B'_{\tau,l} (\mathcal{K}\mu - \Sigma\lambda_0^R) + \frac{1}{2} B'_{\tau,l} \Sigma \Sigma' B_{\tau,l},$$

$$(D-4) \quad \frac{dB_{\tau,l}}{d\tau} = -\rho_1^R - (\mathcal{K} + \Sigma\Lambda^R)' B_{\tau,l},$$

with initial conditions $A_{t,l} = A_t^N$ and $B_{t,l} = B_t^N$. The yield on this artificial indexed bond is:

$$(D-5) \quad \tilde{y}_{t,\tau,l}^I = a_{\tau,l} + b'_{\tau,l} x_t,$$

with factor loadings $a_{\tau,l} = -A_{\tau,l}/\tau$ and $b_{\tau,l} = -B_{\tau,l}/\tau$. In the second step, the yield on the actually marketed indexed bond can be obtained through equation (29) as:

$$(D-6) \quad y_{t,\tau,l}^I = a_{\tau,l} + b'_{\tau,l} x_t - \frac{1}{\tau} \ln \frac{Q_t}{Q_{t-l}}.$$

Finally, to incorporate the TIPS-specific factor, we make use of the 4-factor framework outlined in Appendix C and price the indexed bond using the adjusted real and nominal short rates defined in equations (C-3) through (C-4). In particular, we now use the adjusted nominal short rate defined in equation (C-4) to calculate the time $(T-l)$ price of a l -period nominal bond, $P_{\tau-l,l}^N$, in the first step. This gives us the yield on a τ -year TIPS with an indexation lag l :

$$(D-7) \quad y_{t,\tau,l}^T = y_{t,\tau,l}^R + L_{t,\tau,l} = a_{\tau,l}^T + b_{\tau,l}^{T'} z_t - \frac{1}{\tau} \ln \frac{Q_t}{Q_{t-l}}.$$

Appendix E. State-Space Form

Denote by $x_t = [q_t, \dots, q_{t-l}, x'_t, \tilde{x}_t]'$ the augmented state vector including the natural log of the price level, $q_t \equiv \ln(Q_t)$, its lags over the indexation period, and the TIPS-specific factor, $\tilde{x}_t$. The state equation is derived as the Euler discretization of equations (2), (6), and (24) and can be written in a matrix form as

$$(E-1) \quad x_t = G_h + \Gamma_h x_{t-h} + \eta_t^x,$$

where

$$G_h = \begin{bmatrix} \rho_0^{\pi} h \\ 0 \\ \mathcal{K} \mu h \\ \tilde{\kappa} \tilde{\mu} h \end{bmatrix}, \quad \Gamma_h = \begin{bmatrix} 1 & 0 & \rho_1^{\pi'} h & 0 \\ I_{(l-1) \times (l-1)} & 0 & 0 & 0 \\ 0 & 0 & I_{3 \times 3} - \mathcal{K} h & 0 \\ 0 & 0 & 0 & 1 - \tilde{\kappa} h \end{bmatrix}$$

$$\text{and } \eta_t^x = \begin{bmatrix} \sigma_q' \eta_t + \sigma_q^{\perp} \eta_t^{\perp} \\ 0 \\ \Sigma \eta_t \\ \tilde{\sigma} \tilde{\eta}_t \end{bmatrix}$$

in which η_t , $\eta_t^{\perp}$, and $\tilde{\eta}_t$ are independent of each other and have the distribution

$$(E-2) \quad \eta_t \sim \mathcal{N}(0, h I_{n \times n}), \quad \eta_t^{\perp} \sim \mathcal{N}(0, h), \quad \tilde{\eta}_t \sim \mathcal{N}(0, h).$$

We collect in y_t all data used in the estimation, including the natural log of the current and lagged price levels, $q_{t-i}, i=0, \dots, l$; all nominal yields, $Y_t^N = \{y_{t,\tau_i}^N\}_{i=1}^7$; all TIPS yields, $Y_{t,l}^T = \{y_{t,\tau_i,l}^T\}_{i=1}^3$; short- and long-horizon survey forecasts of future 3-month T-bill yield, $f_t = \{f_{t,6m}, f_{t,12m}, f_{t,long}\}$; and 1- and 10-year-ahead survey inflation forecasts, $\text{IE}_t^{\text{svy}} = \{\text{IE}_{t,1y}^{\text{svy}}, \text{IE}_{t,10y}^{\text{svy}}\}$:

$$(E-3) \quad y_t = [q_t, \dots, q_{t-l}, Y_t^N, Y_{t,l}^T, f_t, \text{IE}_t^{\text{svy}}]'$$

where l is set to 11 weeks. We assume that all nominal and TIPS yields and survey forecasts are observed with error. The observation equation therefore takes the form

$$(E-4) \quad y_t = A + B x_t + e_t,$$

where

$$(E-5) \quad A = \begin{bmatrix} 0 \\ a^N \\ a_l^T \\ a^f \\ a^{IE} \end{bmatrix}, \quad B = \begin{bmatrix} I_{l \times l} & 0_{l \times 3} & 0_{l \times 1} \\ 0_{1 \times l} & b^{N'} & 0 \\ 0_{1 \times l} & b_l^{T'} & \\ 0_{1 \times l} & b^{f'} & 0 \\ 0_{1 \times l} & b^{IE'} & 0 \end{bmatrix}, \quad e_t = \begin{bmatrix} 0 \\ e_t^N \\ e_{t,d}^T \\ e_t^f \\ e_t^{IE} \end{bmatrix},$$

in which the various a vectors and b matrices collect the loadings of nominal and TIPS yields and survey forecasts in obvious ways. We assume a simple independent and identically distributed structure for the measurement errors:

$$(E-6) \quad e_{t,\tau_i}^N \sim \mathcal{N}(0, \delta_{N,\tau_i}^2), \quad e_{t,\tau_i,d}^T \sim \mathcal{N}(0, \delta_{T,\tau_i,d}^2), \quad e_{t,\tau_i}^f \sim \mathcal{N}(0, \delta_{f,\tau_i}^2), \quad e_{t,\tau_i}^{IE} \sim \mathcal{N}(0, \delta_{IE,\tau_i}^2).$$

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